

- 8 D Gravitational force provides for centripetal force.

$$\frac{GMm}{r^2} = mr\omega^2 = mr\left(\frac{2\pi}{T}\right)^2 \Rightarrow \therefore T^2 = r^3$$

Let new period by T' .

$$\left(\frac{T'}{T}\right)^2 = \frac{r^3}{(4r)^3} \Rightarrow T'^2 = 64T^2 \Rightarrow T'^2 = 8T$$

- 9 B $T^2 \propto r^3$

$$\left(\frac{T_2}{T_1}\right)^2 = \left(\frac{r_2}{r_1}\right)^3 \Rightarrow \left(\frac{64T}{T}\right)^2 = \left(\frac{r_2}{r}\right)^3 \Rightarrow r_2 = 16r$$

- 10 A $F = -\frac{\Delta U}{\Delta r} \Rightarrow \Delta U = -F\Delta r$ (Area under F - r graph = change in GPE)

- 11 B $U = 2\left(\frac{-GMm}{r}\right) = 2\left(\frac{-(6.67 \times 10^{-11})(8000)(5)}{0.25}\right) = -2.1 \times 10^{-5} \text{ J}$

- 12 C By definition, gravitational potential is **zero** at infinity and **negative** near the gravitational mass.

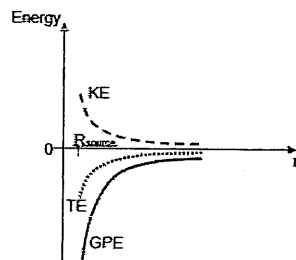
As gravitational field strength is related to the gravitational potential by $g = -\frac{d\phi}{dr}$ (which is the negative of the gradient of the graph of gravitational potential against distance from the centre of the Earth/Moon), the gradient of this graph decreases as the gravitational field strength decreases when the distance from the centre of the Earth increases.

At certain point between the Earth and the Moon, gradient of the graph is zero as the resultant gravitational field strength acting on a mass along XY is zero since the gravitational forces acting on a mass by the Earth and the Moon at that point are opposite in direction but have the same magnitude.

Beyond this point in the direction of the Moon, the gradient of this graph decreases as gravitational force acting on it by the Moon increases as it moves towards the Moon.

- 13 D $E_p = -\frac{GMm}{r}$
 $E_k = \frac{1}{2} \frac{GMm}{r}$ (Check derivation from notes)
 $E_T = E_p + E_k = -\frac{GMm}{r} + \frac{1}{2} \frac{GMm}{r} = -\frac{1}{2} \frac{GMm}{r}$

When moved to higher orbit, r increases.
Hence E_T increases, E_p increases, E_k decreases.



- 14 C $g_{\text{net}} = -\frac{GM}{x^2} + \frac{Gm}{(d-x)^2}$, hence $|g_{\text{m-surface}}| < |g_{\text{M-surface}}|$

There will be a point between the 2 masses where $|g_m| = |g_M|$ and are in opposite direction, hence $g_{\text{net}} = 0$
Since mass M is larger, the point where $g_{\text{net}} = 0$ is closer to mass m .

- 15 D Gravitational force provides for centripetal force.

On moon: $m\left(\frac{g}{6}\right) = m\left(\frac{R}{4}\right)\omega_m^2$ — (1)

On Earth: $mg = mR\omega_E^2$ — (2)

$$\frac{(1)}{(2)}: \frac{1}{6} = \frac{1}{4} \frac{\omega_m^2}{\omega_E^2} \Rightarrow \sqrt{\frac{4}{6}} = \frac{\omega_m}{\omega_E} \Rightarrow \sqrt{\frac{4}{6}} = \frac{T_E}{T_m} \Rightarrow T_m = \sqrt{\frac{3}{2}} T$$

- 16 D By definition.

- 17 B $\rho_p = 2\rho_q$, $r_p = \frac{1}{2}r_q$

$$g_p = \frac{GM_p}{r_p^2} = \frac{G\left(\frac{4}{3}\pi r_p^3 \rho_p\right)}{r_p^2} = \frac{4}{3}G\pi r_p \rho_p = 13.4 \text{ N kg}^{-1}$$

$$g_q = \frac{GM_q}{r_q^2} = \frac{4}{3}G\pi r_q \rho_q = \frac{4}{3}G\pi(2r_p)\left(\frac{1}{2}\rho_p\right) = \frac{4}{3}G\pi r_p \rho_p = 13.4 \text{ N kg}^{-1}$$

- 18 A $U = E_p = -\frac{GMm}{r}$
 $K = E_k = \frac{1}{2} \frac{GMm}{r}$ (Check derivation from notes)
 $E = E_p + E_k = -\frac{GMm}{r} + \frac{1}{2} \frac{GMm}{r} = -\frac{1}{2} \frac{GMm}{r}$

- 19 D Gravitational potential energy is a scalar quantity and hence can only be added up. GPE due to Y is $-1.0 \times 10^{-3} \text{ J}$ and GPE due to Z is $-3.0 \times 10^{-4} \text{ J}$. Hence, the sum of the GPE must be $-1.3 \times 10^{-3} \text{ J}$. This eliminates options A and C.

Gravitational force is a vector quantity. The force due to Y alone is $1.0 \times 10^{-3} \text{ N}$ and the force due to Z alone is $1.5 \times 10^{-4} \text{ N}$. The minimum value obtainable is around $8.5 \times 10^{-4} \text{ N}$. This is if XYZ are on a straight line. The maximum value obtainable is $1.15 \times 10^{-3} \text{ N}$. However, this also requires XYZ in a straight line, and Y and Z to be on the same side of X. This is impossible as XYZ is stated to form a triangle. This eliminates B.

The answer for the force must be a value between $8.5 \times 10^{-4} \text{ N}$ and $1.15 \times 10^{-3} \text{ N}$ exclusive of the limiting values. D fulfils this condition.

- 20 B The work done to bring a 3.0 kg mass from A to infinity is $1000\text{J} + (-60\text{J}) + 50\text{J} = 990\text{J}$. (Conservative force)
The gravitational potential energy is the work done to bring the 3.0 kg mass from infinity to point A, -990J .
The gravitational potential is the work done per unit mass to bring a mass from infinity to point A, -330J .
Hence, the answer is B.

21 D

$$g = \frac{GM}{r^2} = \frac{G(\frac{4}{3}\pi r^3 \rho)}{r^2} = \frac{4}{3}G\rho r$$

hence $g \propto r$ (since they have same density)

- 22 B Potential and g -field caused by P at distance d away.

- 23 A Gravitational force provides for centripetal force.

$$F_g = F_c \Rightarrow \frac{GMm}{r^2} = mr\omega^2 \Rightarrow GM = r^3\omega^2 \text{ (independent of } m\text{)}$$

Geostationary satellites orbit at a fixed position above the Earth's equator. In order to have an orbital period of one day, it can orbit only at a fixed height above earth to satisfy the above equation. $\therefore T, \omega, r$ constant.

$$a_c = r\omega^2 = \text{constant}$$

$\therefore \text{KE} = \frac{1}{2}mv^2$, geostationary satellites can have different kinetic energies.

- 24 A I is a factor because amount of KE required depends on mass of rocket.
II is a factor as the direction of launch when against direction of rotation of Mars implied more energy is required.
III is a factor as different geographic positions may mean different radius of circular motion while Mars is orbiting and hence different initial KE that rocket possess due to rotation of Mars.
IV is a factor as this affects the work done against air resistance.

- 25 C A is wrong because GPE increases to zero.
B is wrong because there is work done against atmosphere.
C is correct because GPE is inversely proportional to distance.
D is wrong because gravitational potential energy is always negative, but changes in this case is positive.

26 A

$$\frac{GMm}{r^2} = mr\omega^2 \Rightarrow \omega^2 = \frac{GM}{r^3}$$

r increases $\Rightarrow \omega$ decreases.

$$E_{\text{total}} = -\frac{GMm}{2r}$$

r increases $\Rightarrow E_{\text{total}}$ gets less negative $\Rightarrow$ i.e.. E_{total} increases.

- 27 C The spaceship's velocity will decrease until zero momentarily at a particular point and then changes direction of motion towards the planet. Hence, v should change signs. So A and B is not plausible. The acceleration of the spaceship should be changing also, so D is definitely not correct.

- 28 C Gravitational force is a conservative force, i.e. it is path independent.

$$\phi_P = -\frac{GM}{r} = -800 \text{ kJ kg}^{-1}$$

$$\phi_Q = -\frac{GM}{2r} = \frac{1}{2}\phi_P = -400 \text{ kJ kg}^{-1}$$

$$\text{Work done by external agent} = U_Q - U_P = m(\phi_Q - \phi_P) = (1)[-400 - (-800)] = 400 \text{ kJ kg}^{-1}$$

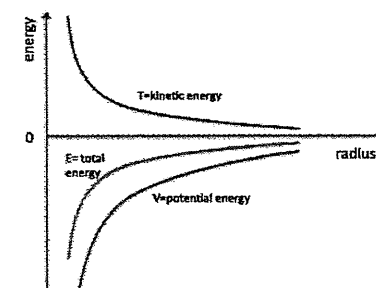
- 29 D For objects in stable orbit about a planet:

P is at a lower orbital radius.

From the graph, P will have higher KE $\Rightarrow$ higher speed.
Hence A and B incorrect.

For smaller r , potential energy is more negative.

Hence C is incorrect.



- 30 D gravitational force on planet $F_g = \frac{GMm}{R^2}$

$$\text{gravitational acceleration} = \frac{F_g}{m} = \frac{GM}{R^2}$$

- 31 D There is only one force, the gravitational force F_g , exerted by the Earth on the satellite which provides the centripetal force required to maintain the satellite in orbit. The centripetal force is a resultant force and it should not be shown in the diagram.

- 32 B Gravitational field strength, $g = \text{constant}$.

$$\text{Change in GPE: } \Delta U = mg\Delta h = m\Delta\phi \Rightarrow g = \frac{\Delta\phi}{\Delta h} = \frac{6}{20}$$

$$\Delta U' = mg\Delta h' = 2\left(\frac{6}{20}\right)(5) = 3.0 \text{ J}$$

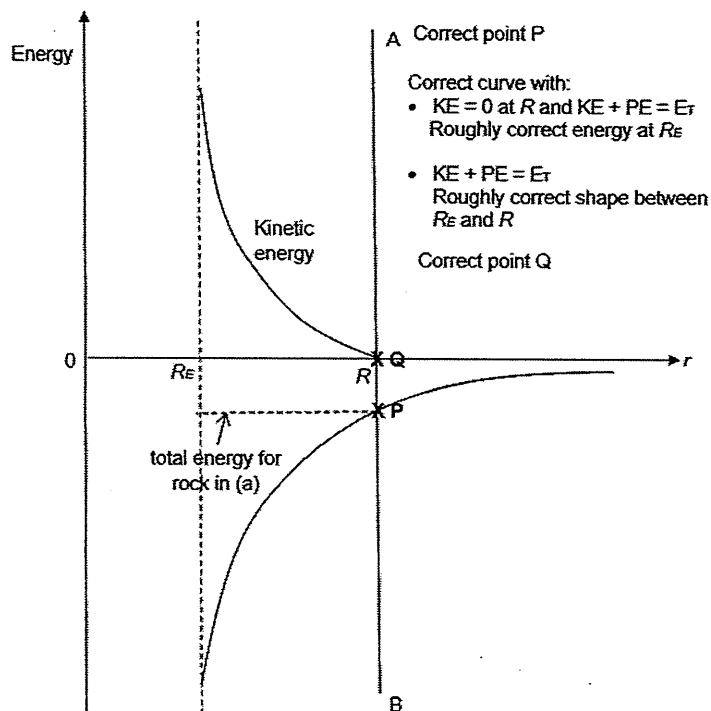
- 33 B Gravitational force provides for centripetal force.

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$v = \sqrt{\frac{GM}{r}} = \sqrt{\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})}{7.55 \times 10^6}} = 7.280 \times 10^3 \text{ m s}^{-1}$$

1. [AJC/Prelims 2014/P2/3]

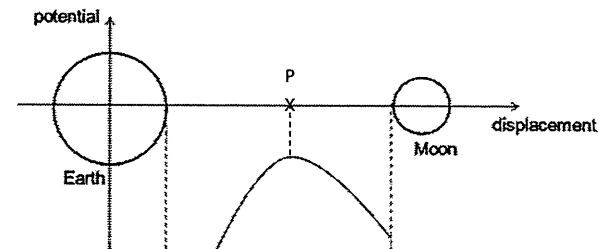
- a. i.
ii.
b. i.



- ii.
- Moon would attract the rock
 - potential at Moon('s surface) is
 - not zero
 - less than 0
 - lower than at infinity
 - moon is nearer; less gain in GPE from Earth to moon

2. [CJC/Prelims 2014/P3/1]

- a. See definitions in lecture notes.
b.



The point P must be marked at the displacement corresponding to the turning point of the graph.

At the turning point, the gradient of the graph, which is the magnitude of the gravitational field strength, is zero. Hence, the net force per unit mass at that point is zero. Therefore, an object at that point will experience no resultant force.

- c. i. At the neutral point, the magnitude of the gravitational field strength due to Earth is equal to the magnitude of the gravitational field strength due to the Moon.

$$g_E = g_M \Rightarrow \frac{GM_E}{x^2} = \frac{GM_M}{y^2} \Rightarrow \frac{x}{y} = \sqrt{\frac{M_E}{M_M}}$$

- ii.

$$\frac{x}{y} = \sqrt{\frac{M_E}{M_M}}$$

$$\frac{x}{3.84 \times 10^5 - x} = \sqrt{\frac{5.97 \times 10^{24}}{7.40 \times 10^{22}}}$$

$$x = 3.45 \times 10^5 \text{ km}$$

- d. At the neutral point, an object experiences no resultant force.
However, an orbiting object requires a non-zero resultant gravitational force to provide for the centripetal force.

3. [DHS/Prelims 2014/P3/6]

- a. See definitions in lecture notes.
b. gravitational force provides centripetal force

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \Rightarrow v = \sqrt{\frac{GM}{r}}$$

- c. See definitions in lecture notes.
d. i. increase in kinetic energy = loss of gravitational potential energy

$$\frac{1}{2}mv_0^2 = 0 - \left(-\frac{GMm}{x}\right)$$

$$v_0 = \sqrt{\frac{2GM}{x}}$$

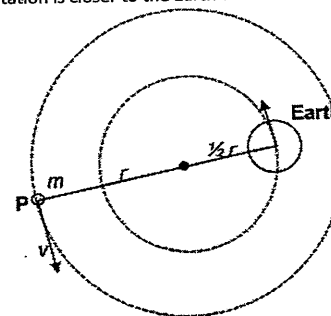
- ii. v_0 is always greater than v for $x = r < \infty$
so stone cannot enter orbit

- e. i. lifting bag experiences upward vertical force due to difference in pressure exerted by surrounding air at the top and bottom surfaces of the lifting bag
upward force is larger than the weight of weather balloon hence upward resultant force.
- ii. magnitude of upwards upthrust = downwards weight
- iii. $g = \frac{GM_{\text{Earth}}}{r^2} \Rightarrow gr^2 = \text{constant}$
 $g_{\text{surface}} R_E^2 = g_{\text{height}} (R_E + 40000)^2$
 $\frac{g_{\text{height}}}{g_{\text{surface}}} \times 100\% = \left(\frac{R_E}{R_E + 40000}\right)^2 \times 100\% = \left(\frac{6400}{6400 + 40000}\right)^2 \times 100\% = 98.8\%$
 Deviation = $100\% - 98.8\% = 1.2\%$
- iv. $T = mg_{\text{height}} \Rightarrow m = \frac{T}{g_{\text{height}}} = \frac{2.42}{(0.988)(9.81)} = 0.250 \text{ kg}$
- v. 1. $s = ut + \frac{1}{2}at^2 \Rightarrow t = \sqrt{\frac{2s}{g}} = \sqrt{\frac{2(40000)}{g}} = 90.8 \text{ s}$
 2. over-estimate, g increases nearer to surface
 (technical definition of "free fall" excludes all other forces except weight)
- f. volume of lifting bag expands on rise due to less atmospheric pressure
 larger initial volume will result in maximum volume and subsequent breakage being reached at a lower height

4. [HCI/Prelims 2014/P3/5]

- a. i. See definitions in lecture notes.
- ii. $\omega = \frac{d\theta}{dt} = \frac{2\pi}{365 \times 24 \times 60 \times 60} = 1.99 \times 10^{-7} \text{ rad s}^{-1}$
- b. i. $F_{\text{Earth}} = \frac{GMm}{r^2} = \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})}{(1.5 \times 10^9)^2} = 1.10 \text{ N}$
 $F_{\text{Sun}} = \frac{GMm}{r^2} = \frac{(6.67 \times 10^{-11})(1.99 \times 10^{30})(6200)}{(1.50 \times 10^9)^2 + (1.496 \times 10^{11})^2} = 36.0 \text{ N}$
 Resultant force = gravitational force by earth + gravitational force by sun = $1.10 + 36.0 = 37.1 \text{ N}$
 Direction: towards the Sun or Earth
- ii. $F_{\text{net}} = mr\omega^2 \Rightarrow 37.1 = (6200)(1.511 \times 10^{11})\omega^2 \Rightarrow \omega = 1.99 \times 10^{-7} \text{ rad s}^{-1}$
- iii. The satellite remains at a distance of $1.5 \times 10^9 \text{ m}$ from the earth, on the line joining the centres of the Earth and Sun. (The satellite, Earth and the Sun are collinear).
- c. i. $\frac{GMm}{r^2} = m \frac{v^2}{r} \Rightarrow v = \sqrt{\frac{GM}{r}} = \sqrt{GM} r^{-\frac{1}{2}}$
 $k = \sqrt{GM}, n = -0.5$
- ii. $\frac{GMm}{r^2} = m \frac{v^2}{r} \Rightarrow \text{Total KE} = \frac{1}{2}mv^2 = \frac{GMm}{2r}$
 $E_r = \text{Total KE} + \text{Total GPE} = \frac{GMm}{2r} + \left(-\frac{GMm}{r}\right) = -\frac{GMm}{2r}$

- iii. 1. Total energy must be zero at infinity. Hence the energy supplied is $\frac{GMm}{2r}$
 $\frac{GMm}{2r} = \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})(2200)}{2(6.70 \times 10^8)} = 6.55 \times 10^{10} \text{ J}$
2. Minimum energy supplied = increase in GPE + increase in KE
 $= (GPE_f - GPE_i) + KE_f - KE_i$
 $= -\frac{GMm}{r} - \left(-\frac{GMm}{R_E}\right) + \frac{GMm}{2r} - 0$
 $= \frac{GMm}{R_E} - \frac{GMm}{2r}$
 $= (6.67 \times 10^{-11})(5.98 \times 10^{24})(2200) \left(\frac{1}{6.4 \times 10^8} - \frac{1}{6.7 \times 10^8}\right) = 7.16 \times 10^{10} \text{ J}$
- OR
 Minimum energy supplied
 = Final Total Energy (in orbit) – Initial Total Energy (at rest at surface)
 $= \left(-\frac{GMm}{2r}\right) - \left(-\frac{GMm}{R_E}\right)$
 $= -6.54 \times 10^{10} + \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})(2200)}{6.4 \times 10^8}$
 $= -(6.54 \times 10^{10}) + (1.371 \times 10^{11}) = 7.17 \times 10^{10} \text{ J}$
- iv. The satellite and the earth would now orbit with the same angular velocity about a point along the centre line joining the two, with the satellite and the earth opposite each other.
 The radius of the satellite orbit is twice the radius of the earth's orbit (or any indication that the centre of rotation is closer to the Earth and to the satellite).



5. [JC/Prelims 2014/P2/2]

- a. i. See definitions in lecture notes.

$$\text{ii. } g = -\frac{d\phi}{dx}$$

- b. i. Change in gravitational potential $(\Delta\phi) = \phi_{\text{final}} - \phi_{\text{initial}} = [-62.53 - (-59.73)] \times 10^6 = -2.80 \times 10^6 \text{ J kg}^{-1}$
 Change in potential energy for satellite $\Delta U = m \Delta\phi = (900)(-2.80 \times 10^6) = -2.52 \times 10^9 \text{ J}$
 Therefore, gravitational potential energy lost = $2.52 \times 10^9 \text{ J}$

$$\text{ii. } g = -\frac{\Delta\phi}{\Delta x} = -\frac{[-59.82 - (-59.64) \times 10^6]}{290\,000 - 310\,000} = -9.0 \text{ N kg}^{-1}$$

Negative sign shows that g is acting in the direction of decreasing r (i.e. towards Earth).

Taking just the magnitude, $g = 9.0 \text{ N kg}^{-1}$

6. [JC/Prelims 2014/P3/7]

- a. See definitions in lecture notes.

- b. Consider a small mass
- m
- located at the surface of the Earth.

The distance between the centre of the Earth and the centre of m is R (the radius of the Earth).

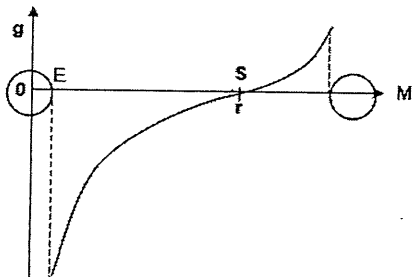
$$\text{gravitational force } F = \frac{GMm}{r^2}$$

$$\text{By definition of gravitational field strength, } g = \frac{F}{m} = \frac{\frac{GMm}{R^2}}{m} = \frac{GM}{R^2}$$

- c. At
- S
- ,
- $g_E - g_M = 0$

$$\frac{GM_E}{r^2} - \frac{GM_M}{(r_0 - r)^2} = 0 \Rightarrow \frac{6.0 \times 10^{24}}{r^2} = \frac{7.0 \times 10^{22}}{(3.8 \times 10^8 - r)^2} \Rightarrow r = 3.4 \times 10^8 \text{ m}$$

- d.



- e. i. Geostationary means that the satellite always appear stationary to an observer on the Earth's surface.
 ii.
 - it must be placed vertically above the equator
 - it must move from west to east
 - the period of revolution of the orbit must be the same as the period of rotation of the Earth about its own axis. i.e. 24 hours

- f. i.

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$\frac{(6.67 \times 10^{-11})(5.97 \times 10^{24})m}{(550 \times 10^3 + 6.38 \times 10^6)^2} = \frac{mv^2}{(550 \times 10^3 + 6.38 \times 10^6)}$$

$$v = 7580 \text{ m s}^{-1}$$

$$\text{ii. } KE = \frac{1}{2}mv^2 = \frac{1}{2}(400)(7580)^2 = 11.5 \times 10^9 \text{ J}$$

$$GPE = -\frac{GMm}{r} = -\frac{(6.67 \times 10^{-11})(5.97 \times 10^{24})(400)}{(550 \times 10^3 + 6.38 \times 10^6)} = -2.30 \times 10^{10} \text{ J}$$

$$\text{total energy of the satellite} = KE \text{ of satellite} + GPE \text{ of satellite} = 11.5 \times 10^9 - 2.30 \times 10^{10} = -1.15 \times 10^{10} \text{ J}$$

OR

Total energy of the satellite

$$= KE \text{ of satellite} + GPE \text{ of satellite} = \left(\frac{GMm}{2r}\right) + \left(-\frac{GMm}{r}\right)$$

$$= -\frac{GMm}{2r} = -\frac{(6.67 \times 10^{-11})(5.97 \times 10^{24})(400)}{2(550 \times 10^3 + 6.38 \times 10^6)} = -1.15 \times 10^{10} \text{ J}$$

$$\text{iii. } v = r\omega \Rightarrow 7580 = (550 \times 10^3 + 6.38 \times 10^6)\left(\frac{2\pi}{T}\right) \Rightarrow T = 5744 \text{ s} = 1.60 \text{ hr}$$

Since the period of satellite is less than 24 hours, it is NOT geostationary.

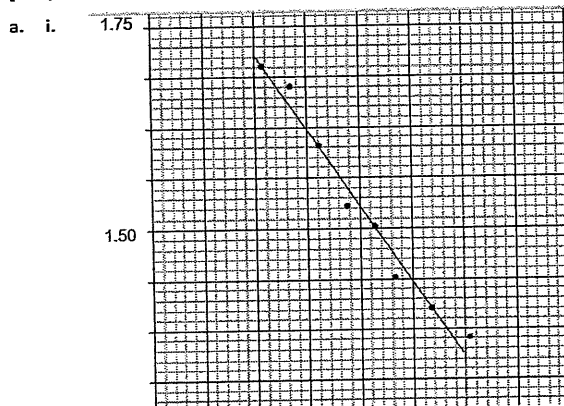
OR

$$r^3 = \frac{GM}{\left(\frac{2\pi}{T}\right)^2} = \frac{(6.67 \times 10^{-11})(5.97 \times 10^{24})}{\left(\frac{2\pi}{24 \times 3600}\right)^2} \Rightarrow r = 4.22 \times 10^7 \text{ m}$$

$$\text{radius of orbit} = (550 \times 10^3) + (6.38 \times 10^6) = 6.93 \times 10^6 \text{ m} \neq r.$$

- iv. A satellite in a geostationary orbit is often used for telecommunications because it will always remain at the same location, in constant electronic line-of-sight to the data receiving stations on the Earth's surface, which facilitates telecommunications.

7. [MJC/Prelims 2014/P2/2]



ii. Gradient = -0.94 ± 0.02

Y-intercept = 1.80 ± 0.02

$$R/R_s = -0.94 (M/M_s) + 1.80$$

iii. 1.92 ± 0.02

iv. This value is obtained by extrapolation outside the data range and hence uncertainties are present.

Or The trend may not be a straight line beyond the set of data points given.

Or The maximum ratio corresponds to a star with zero radius hence this is not possible in reality.

v. No, the Sun does not belong to this species of star.

The data point which represents the Sun on the graph is found at coordinates (1,1), which does not lie in the trend presented by the data points.

b. i. Change in gravitational potential = $-4.2 - (-13.5) = 9.3 \times 10^6 \text{ J kg}^{-1}$

ii. Energy at surface = Energy at 3R above surface

$$KE_R + PE_R = KE_{4R} + PE_{4R}$$

$$\frac{1}{2}mv^2 + m\phi_R = 0 + m\phi_{4R}$$

$$\frac{1}{2}v^2 = -4.2 \times 10^6 - (-13.5 \times 10^6)$$

$$v = 4.3 \times 10^3 \text{ m s}^{-1}$$

iii. $g = \frac{d\phi}{dr} = \frac{\Delta\phi}{\Delta r} = \frac{[-3.8 - (-4.6)] \times 10^6}{(4.5 - 3.5) \times 3300 \times 10^3} = 0.24 \text{ N kg}^{-1}$

iv. The claim is valid.

The gravitational field strength at any point is a resultant of all the gravitational field strengths of this star and its neighbouring stars.

8. [NJC/Prelims 2014/P3/6]*

a. i. -12.6 MJ of work done by external agent is needed to move a unit mass from infinity to the point.

ii. Use of ratios (inverse r law attempt), -6.32 MJ kg^{-1}

b. i. No change in gravitational PE/still on same equipotential
Or No work done moving along the equipotential surface

ii. KE At D = 1.143 GJ (Allow substitution in formula)

Change in gravitational PE = $850 \times 1.04 \text{ MJ} = 0.884 \text{ GJ}$

Total energy at B = $1.143 + 0.884 \text{ (GJ)} = 2.027 \text{ GJ}$

Speed at B = 2190 m s^{-1}

c. i. Use of period = $24.6 \times 60 \times 60 (8.86 \times 10^4 \text{ s})$ or $\omega = 7.09 \times 10^{-5} (\text{rad s}^{-1})$
 $R = 2.04 \times 10^7 \text{ m} (20\,400 \text{ km})$

ii. E_t becomes more negative, radius of orbit will decrease as magnitude of E_t will increase as radius decrease.
Since $E_t = -E_k$ and E_t becomes more negative, E_k will increase and hence speed of satellite v will increase.

d. i. volume = mass/density = $(1.4 / 810) = 0.00173 \text{ m}^3$

ii. $pV = nRT \Rightarrow 1.0 \times 10^5 \times V = \left(\frac{1.4}{0.0280}\right) \times 8.31 \times (25 + 273.15) \Rightarrow V = 1.24 \text{ m}^3$

iii. In the absence of external forces, the total momentum of the system is conserved. As the initial total momentum of the system is zero, by Principle of conservation of linear momentum, when the mass of gases is ejected backwards through the nozzle (its momentum changes), there will be an equal and opposite momentum gained by the astronaut, so that the final total momentum will be zero.

The rate of change of momentum of the astronaut in the new direction will result in a forward force on the astronaut based on Newton's 2nd law.

iv. As the volume of the gas expand, work is done by the gas. Hence the work done on the gas (W) is negative. There is no time for heat transfer during the gas expansion; hence the heat supplied to the gas (Q) is zero. By first law of thermodynamics, the increase in internal energy of the gas will be negative as given by $\Delta U = Q + W$ where Q is zero and W is negative.

Therefore, there is a decrease in the internal energy of the gas.

9. [NYJC/Prelims 2014/P3/1]

a. i. See definitions in lecture notes.

ii. See definitions in lecture notes.

b. i. Loss in potential energy = $-m\Delta\phi = -700[(-62.72) - (-59.03)] \times 10^9 = 2.58 \times 10^9 \text{ J}$

ii. Since $g = -\frac{d\phi}{dr}$, to deduce g at 400000 m ,

calculate the gradient of it using values at 390000 m and 410000 m

$$|g| = \left| \frac{\Delta\phi}{\Delta r} \right| = \left| \frac{[(-59.12) - (-58.94)] \times 10^9}{390000 - 410000} \right| = 9.0 \text{ N kg}^{-1}$$

10. [PJC/Prelims 2014/P3/6]

a. $F_g = \frac{Gm_1m_2}{r^2}$

where F_g is the gravitational force between two masses m_1 and m_2

G is the gravitational constant

r is the center-to-center distance between masses m_1 and m_2

- b. i. A geostationary orbit is one in which the object in orbit is always above the same point on the Earth's surface.

- ii. 24 hours, West to East.

- iii. The satellite must rotate in planes perpendicular to the axis of the Earth's rotation in order for it to rotate together with the Earth and maintain a fixed point above it.

The satellite's center of orbit must be the center of the Earth in order for gravitational force alone to act as centripetal force.

Hence, a geostationary satellite is confined to the plane of the equator of the Earth, and will always be above the equator.

c. i. $\omega = \frac{2\pi}{T} = \frac{2\pi}{24(60)(60)} = 7.27 \times 10^{-5} \text{ rad s}^{-1}$

$$\frac{Gm_1M_E}{r^2} = m_s r \omega^2$$

$$GM_E = r^3 \omega^2$$

$$r^3 = \frac{(6.67 \times 10^{-11})(5.97 \times 10^{24})}{(7.27 \times 10^{-5})^2} = 7.53 \times 10^{22}$$

$$r = 4.2 \times 10^7 \text{ m}$$

$$\text{height above Earth} = 4.2 \times 10^7 \text{ m} - 6371000 \text{ m} = 3.6 \times 10^7 \text{ m}$$

ii. $t = \frac{2h}{c} = \frac{2(3.6 \times 10^7)}{3.0 \times 10^8} = 0.239 \text{ s}$

- iii. It will remain the same. For a geostationary satellite, the orbital height is fixed regardless of the mass of the satellite, as the radius is not a function of the mass of the satellite.

OR

As can be seen from earlier calculations, the radius of a satellite depends on the mass of the Earth, the gravitational constant and the orbital period. For a geostationary satellite, the orbital period is a constant, and hence so is the radius of revolution.

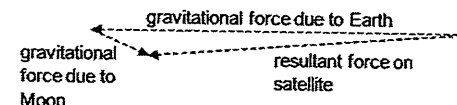
iv. $F_g = \frac{Gm_1m_2}{r^2} = \frac{6.67 \times 10^{-11} \times 5.97 \times 10^{24} \times 100}{(4.2 \times 10^7)^2} = 21.3 \text{ N}$

- d. (scale exaggerated: force due to moon is smaller than that due to Earth. Angle between forces should NOT be labelled θ)

The net force (and hence acceleration) caused by the forces of the moon and the Earth acting on the satellite is at an angle away from the plane of the equator.

If this is not corrected, the satellite will start to deviate from its geostationary orbital path.

Hence, the thrusters are needed to counteract the additional force from the moon. (and also other astronomical bodies.)



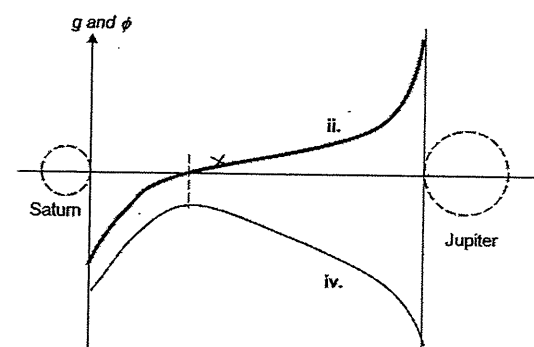
11. [RVHS/Prelims 2014/P3/7]

a. i. $g \text{ due to Jupiter} - g \text{ due to Saturn} = \frac{(6.67 \times 10^{-11})(5.7 \times 10^{28})}{(3.9 \times 10^{11})^2} - \frac{(6.67 \times 10^{-11})(1.9 \times 10^{27})}{(4.2 \times 10^{11})^2} = -4.7 \times 10^{-7}$
 $g = 4.7 \times 10^{-7} \text{ N m (towards Saturn)}$

ii.

iii.

iv.



v. 1. $T = 10.7 \times 3600 = 3.85 \times 10^4 \text{ s}$

2. The gravitational force provides centripetal force.

$$GMm/r^2 = m\omega^2 r$$

$$r = \{(6.67 \times 10^{-11} \times 5.7 \times 10^{26} / 4\pi^2 \times (3.85 \times 10^4)^2)^{1/3}$$

$$r = 1.13 \times 10^8 \text{ m}$$

- vi. Stationary satellite requires it to be above same point on planet, and since g -force must act through the centre of the planet, the only plane that satisfy this condition is the equatorial plane.

- vii. The reading is zero. The acceleration due to gravity in a free orbit is sufficient to provide for the centripetal acceleration, hence the reading acting on the spring balance on the object has to be zero.

(no additional force is needed to provide for that centripetal force).

OR

Object, spring balance and satellite are moving with the same acceleration in that orbit, which is provided by the gravitational force. Hence there is no need for additional force between object and the spring balance.

- b. i. (use any two sets of data to show proportional relationship)

If $T^2 \propto R^3$, then $T^2/R^3 = \text{constant}$

$$T_A^2/R_A^3 = (0.500)^2/(1.81 \times 10^8)^3 = 4.2 \times 10^{-26}$$

$$T_B^2/R_B^3 = (3.55)^2/(6.71 \times 10^8)^3 = 4.2 \times 10^{-26}$$

Since the two values of T^2/R^3 is the same, it is verified.

ii. Constant = $4.2 \times 10^{-26} \text{ days}^2 \text{ m}^{-3}$ or $3.1 \times 10^{-16} \text{ s}^2 \text{ m}^{-3}$

iii. Constant of proportionality = $\frac{4\pi^2}{g_p R_p^2} = 3.1 \times 10^{-16}$

$$g_p = \frac{4\pi^2}{(3.1 \times 10^{-16})(6.9 \times 10^7)^2} = 26.7 \text{ N kg}^{-1}$$

iv. The proportionality relationship $T^2 \propto R^3$ is only used for objects orbiting around the same mass.

If another planet is used, the constant of proportionality will be different and the values from Fig. 11.2 will not be applicable to the Moon orbiting round Earth.

12. [SAJC/Prelims 2014/P3/2]

- a. A geostationary satellite is one which is always above the same location on the Earth's surface as the Earth rotates about its axis.
- b. Near the equator, it has higher linear speed (or KE) and hence less energy needs to be supplied to bring satellite to eventual orbit.

- c. i. Gravitational force provides centripetal force

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \Rightarrow KE = \frac{1}{2}mv^2 = \frac{GMm}{2r}$$

$$\Delta KE = \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})(120)}{2} \left(\frac{1}{3.00 \times 10^7} - \frac{1}{4.23 \times 10^7} \right) = 2.32 \times 10^8 \text{ J}$$

ii. $GPE = -\frac{GMm}{r}$

$$\Delta GPE = -(6.67 \times 10^{-11})(5.98 \times 10^{24})(120) \left(\frac{1}{3.00 \times 10^7} - \frac{1}{4.23 \times 10^7} \right) = -4.64 \times 10^8 \text{ J}$$

- d. Half the GPE lost is converted to kinetic energy, resulting in an increase in speed and half the GPE lost is converted to thermal energy due to air resistance.
- e. The sensation of weight is due to the normal reaction exerted on the object.
The astronaut and the satellite have the same acceleration and therefore, the normal reaction on the astronaut is zero.

13. [SRJC/Prelims 2014/P3/2]

- a. i. Mass of the planet.

ii. $\text{speed} = \frac{\text{circumference}}{\text{time}} = \frac{2\pi r}{T}$

As $T^2 \propto r^3$,

$$\text{speed} = \frac{2\pi r}{\frac{1}{k^2 r^2}} = \frac{2\pi}{\frac{1}{k^2 r^2}}$$

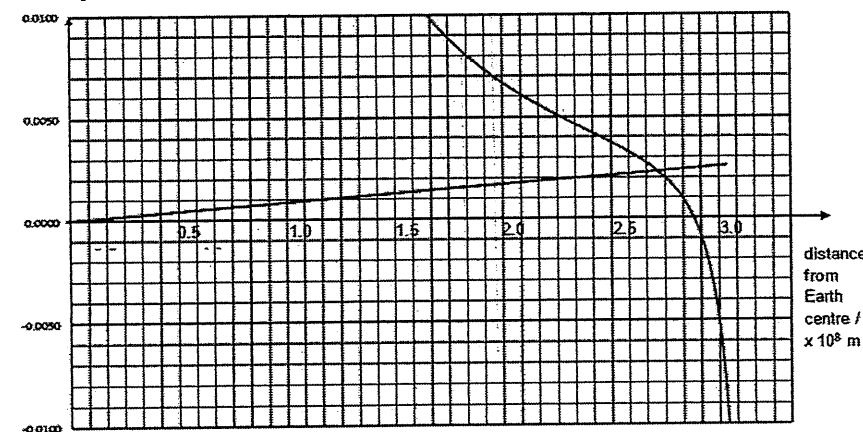
Since kinetic energy is proportional to square of speed, when radius is larger, speed is smaller and hence kinetic energy is lesser.

- b. i. Resultant gravitational force provides centripetal force

$$m(g_r - g_m) = mr\left(\frac{2\pi}{T}\right)^2$$

$$(g_r - g_m) = r\left(\frac{2\pi}{T}\right)^2$$

- ii. net gravitational field strength/ N kg^{-1}



- iii. The line drawn represents centripetal acceleration = $r\left(\frac{2\pi}{T}\right)^2$, where T is the period of the moon.

At the distance where this line intersects the net gravitational field strength line, it implies the net gravitational force provides the centripetal force necessary to have the intended period. and hence, this distance is the radius of the orbital path of the satellite.

14. [TJC/Prelims 2014/P3/5]

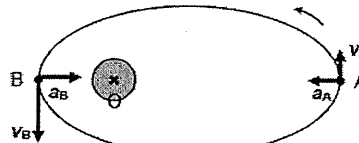
- a. A region in space where a mass placed in it will experience a gravitational force.

distance from centre of planet / 10^5 m	gravitational potential / 10^3 J kg $^{-1}$	$\phi r / 10^8$
1.25 (surface of planet)	-15.0	-18.8
3.00	-6.25	-18.8
7.50	-2.50	-18.8

Since $\phi r = \text{constant}$, this implies ϕ is inversely proportional to r .

- ii. $\phi r = -GM = -18.8 \times 10^8 \Rightarrow GM = 18.8 \times 10^8$
 $g = \frac{GM}{r^2} = \frac{18.8 \times 10^8}{(1.25 \times 10^5)^2} = 0.120 \text{ N kg}^{-1}$
- iii. 1. $F_c = mr\omega^2 = mr\left(\frac{2\pi}{T}\right)^2 = (3.0 \times 10^4)(1.25 \times 10^5)\left(\frac{2\pi}{5.0 \times 60 \times 60}\right)^2 = 457 \text{ N}$
 2. Gravitational force acting at surface on vehicle, $F_g = mg = 3.0 \times 10^4 \times 0.120 = 3600 \text{ N}$
 Since $F_g > F_c$
 $\Rightarrow$ contact force between vehicle and surface of planet > 0 ,
 $\Rightarrow$ vehicle will stay on surface on planet
- c. i. minimum speed of object at planet's surface so that it will escape from the gravitational field / travel to infinity
- ii. Gravitational potential energy at surface of planet $= -\frac{GMm}{r}$
 For the probe to escape to deep space, the total energy of the probe at launch must be more than zero.
 Since the energy given to probe $\left(\frac{3GMm}{4R}\right)$ is less than $\frac{GMm}{r}$, the probe will not be able to escape.
- iii. 1. Change in gravitational PE $= GMm\left(\frac{1}{R} - \frac{1}{r}\right)$
 2. Gravitational force provides centripetal force
 $\frac{GMm}{r^2} = \frac{mv^2}{r} \Rightarrow KE = \frac{1}{2}mv^2 = \frac{GMm}{2r}$
- iv. Energy supplied = increase in KE + increase in GPE
 $\frac{3GMm}{4R} = \frac{GMm}{2r} + \frac{GMm}{R} - \frac{GMm}{r}$
 $\frac{1}{4R} = \frac{1}{2r}$
 $r = 2R$
 Hence, height above the surface $= R$

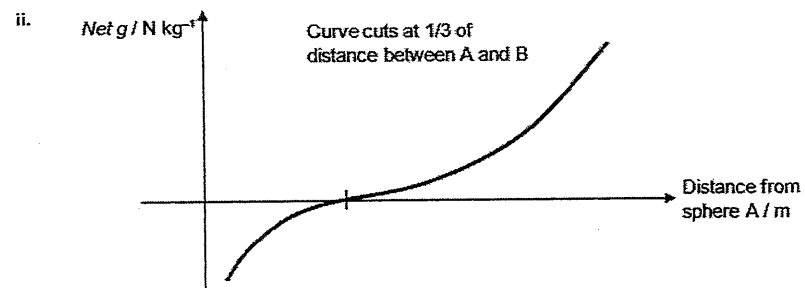
15. [TPJC/Prelims 2014/P3/6]

- a. i. See definitions in lecture notes
- ii. $\omega = \frac{2\pi}{T} = \frac{2\pi}{365 \times 24 \times 60 \times 60} = 1.99 \times 10^{-7} \text{ rad s}^{-1}$
- b. i. 1. Force by Earth $F_E = \frac{GM_E m}{r_{EX}^2} = \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})(425)}{(1.6 \times 10^8)^2} = 0.06621 \text{ N}$
 Force by Sun $F_S = \frac{GM_S m}{r_{SX}^2} = \frac{(6.67 \times 10^{-11})(1.99 \times 10^{30})(425)}{(1.50 \times 10^{11} - 1.6 \times 10^8)^2} = 2.561 \text{ N}$
 Resultant force $= 2.561 - 0.06621 = 2.49 \text{ N}$ towards the Sun
 2. acceleration $= \frac{F}{m} = \frac{2.49}{425} = 5.87 \times 10^{-3} \text{ m s}^{-2}$
- ii. $a = r\omega^2 \Rightarrow \omega = \sqrt{\frac{a}{r}} = \sqrt{\frac{5.87 \times 10^{-3}}{1.50 \times 10^{11} - 1.60 \times 10^8}} = 1.99 \times 10^{-7} \text{ rad s}^{-1}$
- iii. Since angular velocity of satellite is the same as that of the Earth, the satellite will remain in the same position relative to the Earth as it goes round the Sun.
- iv. This orbit is preferable because there will never be an instance when the Earth will come between the satellite and the Sun, and thus the satellite will be able to observe the Sun continuously.
- v. Possible disadvantages:
 As the satellite is between the Earth and the Sun, the satellite needs to have a very powerful transmitter to allow its signal to be strong enough to be distinguished from the electromagnetic waves emitted by the Sun.
 This orbit is unstable and any disturbance (solar flare, solar radiation pressure, impact with space dust) will cause the satellite to move from this orbit. The satellite will need a large fuel tank to keep it in this orbit.
 It is more expensive and more difficult to send a satellite to this orbit as compared to an orbit around the Earth because of the long distance.
- Fun facts:
 The only satellite that can be repaired in space is the Hubble Space Telescope as it is in low earth orbit and is designed to be serviced by the Space Shuttle. As the Space Shuttle program ended in 2011, there is currently no ability to repair satellites in orbit.
 The Deep Space Network is NASA's international array of giant radio antennas that supports interplanetary spacecraft missions, plus a few that orbit Earth.
- c. i. 1. 
 2.
- ii. 1. A is further away, so GPE is higher. As the total mechanical energy in an orbit is constant, the KE, and therefore the velocity at A will be lower. v_A is drawn shorter
 2. A is further away, so the gravitational force is weaker, and therefore acceleration is smaller. a_A is drawn shorter.

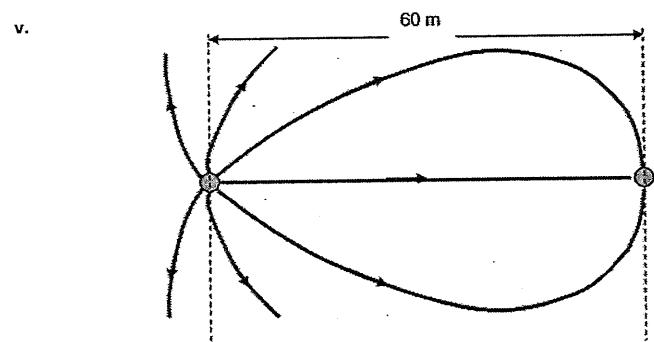
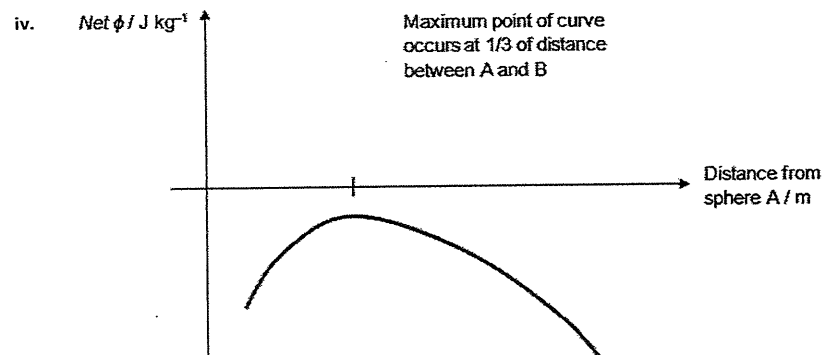
16. [YJC/Prelims 2014/P3/7]*

a. See definitions in lecture notes

- b. i. g by sphere A = g by sphere B
 $G(100)/x^2 = G(400)/(60-x)^2$
 $x = 20 \text{ m}$



iii. $g = -\frac{d\phi}{dr}$



vi. $\text{Net } E = E$ by sphere A + E by sphere B = $\frac{10}{4\pi\epsilon_0(20)^2} + \frac{5}{4\pi\epsilon_0(40)^2} = 2.53 \times 10^8 \text{ N C}^{-1}$

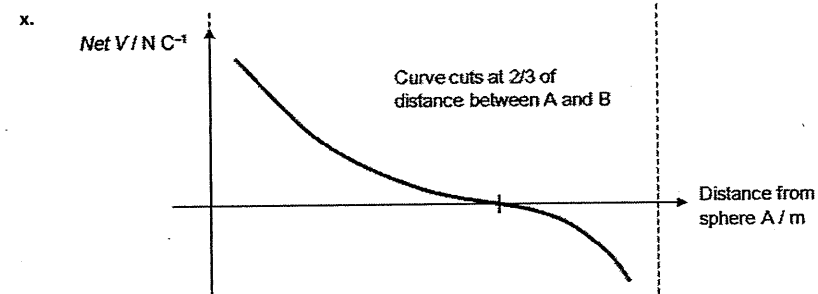
vii. C will accelerate towards B due to the net electric force since net gravitational force is zero.

viii. Work done per unit positive charge in bringing a point charge from infinity to that point.

ix. V of sphere A = V of sphere B

$$\frac{10}{4\pi\epsilon_0 y} = \frac{5}{4\pi\epsilon_0(60-y)}$$

$$y = 40 \text{ m}$$

xi. The net E will not be zero as the E due to spheres A and B are both pointing to the right.

Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs

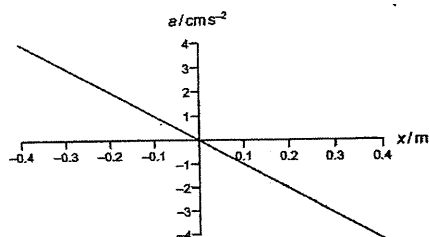
1. [ACJC/Prelims 2014/P1/17]

Simple harmonic motion is defined as the motion of a particle such that

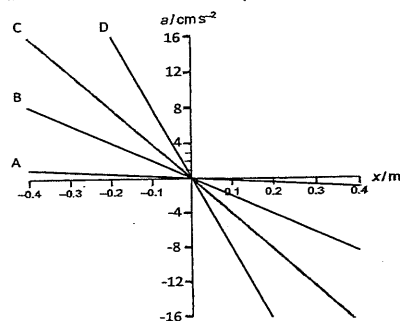
- A its displacement x from the equilibrium position is always given by the expression $x = x_0 \sin \omega t$.
- B its displacement x from the equilibrium position is related to its velocity by the expression $V = \omega x$.
- C its acceleration is proportional to, and in the opposite direction to, the displacement from the equilibrium position.
- D Its acceleration is always $\omega^2 x_0$ and is directed at right angles to its motion.

2. [ACJC/Prelims 2014/P1/18]

The graph shows the variation of acceleration a with displacement x for an object undergoing simple harmonic motion.



The simple harmonic motion system is altered so that it has a period half of that before.
Which line shows the new variation of acceleration a with displacement x for the object?



3. [ACJC/Prelims 2014/P1/19]

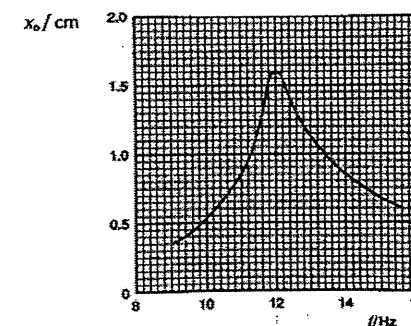
A particle is in simple harmonic motion.

What is the magnitude of the phase difference between the velocity and displacement of particle?

- A $\frac{\pi}{4}$ rad
- B $\frac{\pi}{2}$ rad
- C π rad
- D $\frac{3\pi}{2}$ rad

4. [ACJC/Prelims 2014/P1/20]

The variation with frequency f of the amplitude x_0 of the forced oscillations of a machine is as shown.

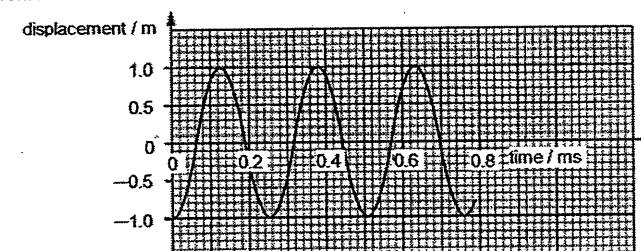


Which statement is **not** correct?

- A Resonance frequency of the above system is 12 Hz.
- B For no damping, the maximum amplitude is 1.6 cm.
- C Increased damping reduces the sharpness of the resonance peak.
- D Damping decreases the maximum amplitude of the oscillations.

5. [AJC/Prelims 2014/P1/15]

A body moves with simple harmonic motion about a point P. The graph shows the variation with time of its displacement from P.

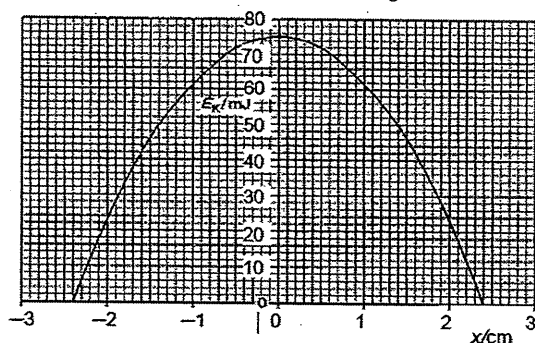


What is its angular frequency?

- A $1.6 \times 10^{-3} \text{ rad s}^{-1}$
- B $4.0 \times 10^{-3} \text{ rad s}^{-1}$
- C $4.0 \times 10^3 \text{ rad s}^{-1}$
- D $2.5 \times 10^4 \text{ rad s}^{-1}$

6. [AJC/Prelims 2014/P1/16]

Two horizontal springs are attached to a trolley. The trolley is then displaced along the axis of the springs and then released. The trolley undergoes simple harmonic motion. The variation of the kinetic energy E_k of the trolley with horizontal displacement x from its equilibrium position is shown in the figure below.



The trolley loses energy so that its maximum kinetic energy is reduced by 50 mJ.

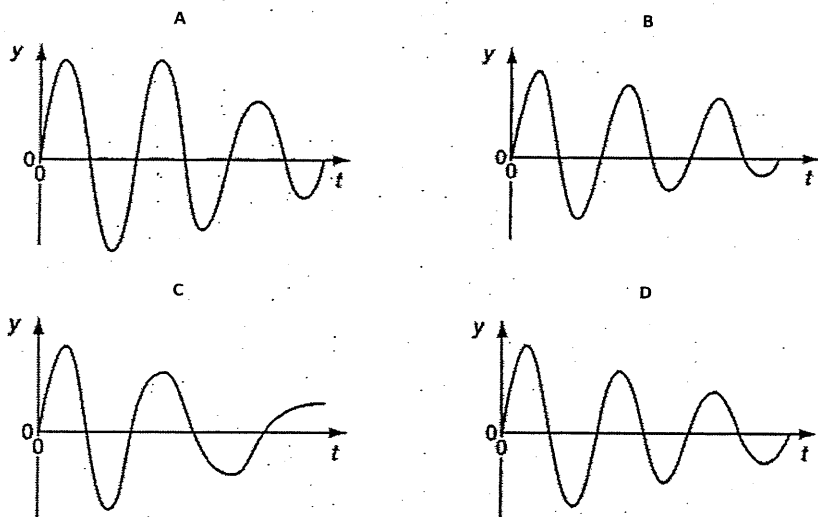
What is the amplitude of the oscillations?

- A 2.4 cm B 1.9 cm C 1.4 cm D 0.95 cm

7. [CJC/Prelims 2014/P1/15]

A mass-spring system is placed in water and allowed to oscillate with decreasing amplitude until it comes to a complete stop.

Which diagram shows the variation of the displacement y with time t of the system?



8. [CJC/Prelims 2014/P1/16]

A block P, of mass m , attached to a spring with spring constant k , oscillates in simple harmonic motion on a frictionless horizontal surface. A block Q, of the same mass as P, is attached to a similar spring and oscillates vertically in simple harmonic motion. The angular frequency ω of the oscillation is determined by $\omega = \sqrt{\frac{k}{m}}$.

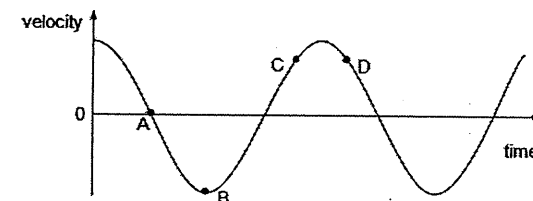
Given that both blocks oscillate with the same amplitude, which of the following statements is true?

- A The maximum kinetic energy of P is smaller than the maximum kinetic energy of Q.
 B The maximum kinetic energy of P is larger than the maximum kinetic energy of Q.
 C The maximum elastic potential energy of P is smaller than the maximum elastic potential energy of Q.
 D The maximum elastic potential energy of P is larger than the maximum elastic potential energy of Q.

9. [DHS/Prelims 2014/P1/15]

The diagram shows the graph of velocity against time for a body performing simple harmonic motion.

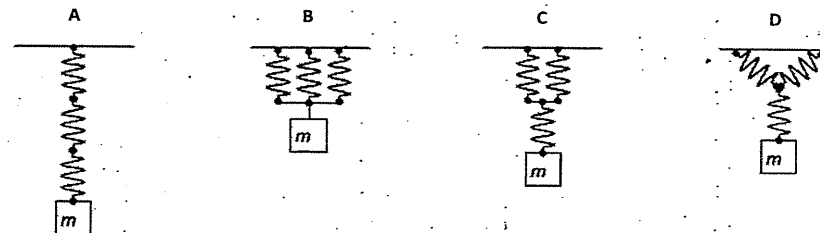
At which point are the velocity and acceleration in opposite directions?



10. [DHS/Prelims 2014/P1/16]

The diagram below shows 3 identical springs arranged in different manners.

Which arrangement will result in vertical oscillations of the highest frequency?



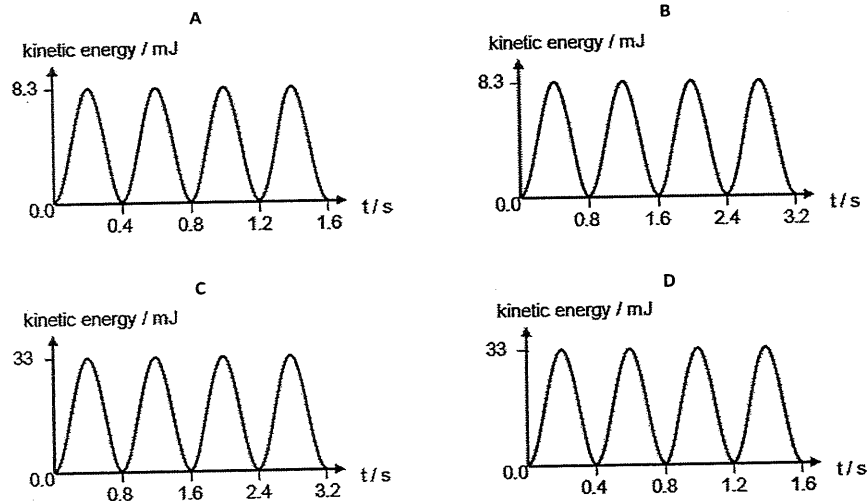
11. [HCI/Prelims 2014/P1/15]

A rubber ducky of mass 300 g is floating in a fish tank. When it is pushed into the water without fully submerging it and released, it bobs up and down. The distance of the rubber ducky from the base of the tank is given by

$$d = 0.25 - 0.03 \cos 2.5\pi t$$

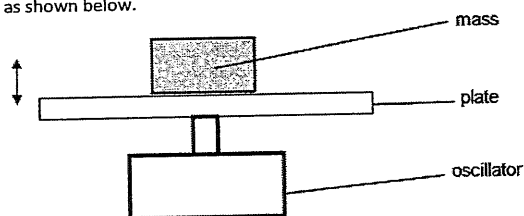
where d is given in metres and t is the time in seconds.

Which of the following graphs shows the variation with time of the rubber ducky's kinetic energy?



12. [HCI/Prelims 2014/P1/16]

A 0.50 kg mass is placed on a flat horizontal plate and the plate is made to oscillate in the vertical direction with simple harmonic motion as shown below.



The plate oscillates with a frequency of 13 Hz.

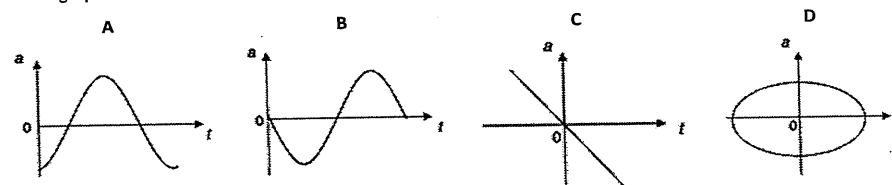
What is the amplitude of oscillation of the plate so that the mass just loses contact with the plate?

- A 0.74 mm B 1.5 mm C 6.1 mm D 12 mm

13. [UC/Prelims 2014/P1/15]

A pendulum moving with simple harmonic motion has maximum potential at time $t = 0$ s.

Which graph best describes the variation of the acceleration a of the pendulum with time t ?



14. [UC/Prelims 2014/P1/16]

When an object is displaced from its equilibrium position, the oscillating system undergoes critical damping.

Which of the following is true about a critically damped system?

- A The object oscillates with its amplitude varying linearly with time.
 B The object does not return to its equilibrium position for a very long time.
 C The object returns to its equilibrium position in the shortest possible time.
 D The object oscillates with its amplitude decreasing exponentially with time.

15. [MJC/Prelims 2014/P1/15]

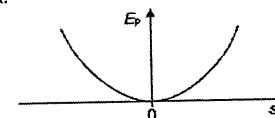
A housewife notices that her washing machine vibrates most vigorously whenever the r.p.m. (revolutions per minute) of the spinning drum reaches a certain value. She sends it for repair and the repairman merely attached a heavy mass to the drum of the washing machine.

Which of the following outcomes is most likely to happen to the washing machine?

- A It does not vibrate vigorously anymore at all r.p.m. values.
 B It vibrates most vigorously at the same r.p.m. value.
 C It vibrates most vigorously at a higher r.p.m. value.
 D It vibrates most vigorously at a lower r.p.m. value.

16. [MJC/Prelims 2014/P1/16]

The graph below shows the variation of the potential energy E_p of a body with its displacement s from a fixed point X.

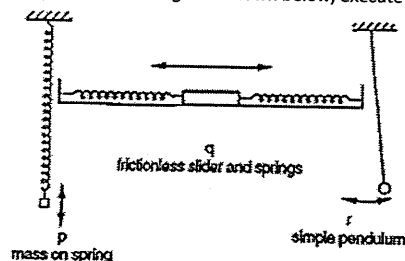


Which feature of the graph best indicates that the body is undergoing simple harmonic motion?

- A The graph passes through the origin.
 B The potential energy does not have negative values.
 C The graph is approximately linear for large values of s .
 D The value of the potential energy increases as the body moves away from point X.

17. [NJC/Prelims 2014/P1/17]

All three systems, represented as p, q and r in the diagrams shown below, execute simple harmonic system.



In which system will the period be independent of the mass of the body?

- A p only B q only C r only D p, q and r

18. [NYJC/Prelims 2014/P1/15]

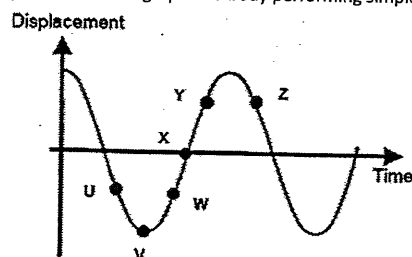
A sphere attached to a horizontal spring is oscillating on a smooth horizontal surface at a frequency of 2.0 Hz. The amplitude of oscillation is 2.0 cm.

What is the speed of the sphere when it is at a distance 1.0 cm from equilibrium position?

- A 6.3 cm s^{-1} B 22 cm s^{-1} C 25 cm s^{-1} D 31 cm s^{-1}

19. [NYJC/Prelims 2014/P1/16]

The diagram below shows a displacement-time graph of a body performing simple harmonic motion.



At which one of the points, U, V, W, X, Y or Z, is the body travelling and accelerating in the opposite direction?

- A U, Y B V, X C W, Z D X, Z

20. [PJC/Prelims 2014/P1/15]

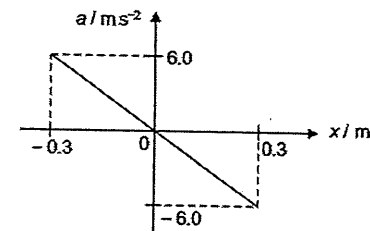
A mass of 0.3 kg is placed on a piston oscillating vertically in simple harmonic motion with a period of 0.50 s. The maximum displacement of the piston from its equilibrium position is 2.0 cm.

What is the minimum force between the piston and the mass during the oscillation?

- A 0 N B 2.0 N C 2.9 N D 3.9 N

21. [PJC/Prelims 2014/P1/16]

A particle moves such that its acceleration a is related to its displacement x from a fixed point as shown below.



What is the magnitude of its maximum velocity?

- A 0.40 m s^{-1} B 1.3 m s^{-1} C 1.8 m s^{-1} D 30 m s^{-1}

22. [RVHS/Prelims 2014/P1/13]

A mass of 2.0 kg attached to a light spring is oscillating on a smooth horizontal surface at a frequency of 1.0 Hz. The other end of the spring is fixed to a wall. The amplitude of oscillation of the spring-mass system is 4.0 cm.

What is the speed of the mass when it is at a distance 3.0 cm from its equilibrium position?

- A 2.6 cm s^{-1} B 3.9 cm s^{-1} C 17 cm s^{-1} D 24 cm s^{-1}

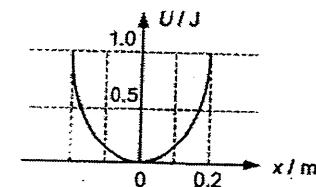
23. [RVHS/Prelims 2014/P1/14]

When damping is increased on a mass undergoing oscillation at its resonant frequency, which effect does **not** occur?

- A The resonant frequency decreases.
B The sharpness of the resonant peak decreases.
C The amplitude of the forced oscillation decreases.
D The period of the forced oscillation at resonance decreases.

24. [SAJC/Prelims 2014/P1/17]

A particle of mass 4 kg moves with simple harmonic motion and its potential energy U varies with position x as shown in the figure below.



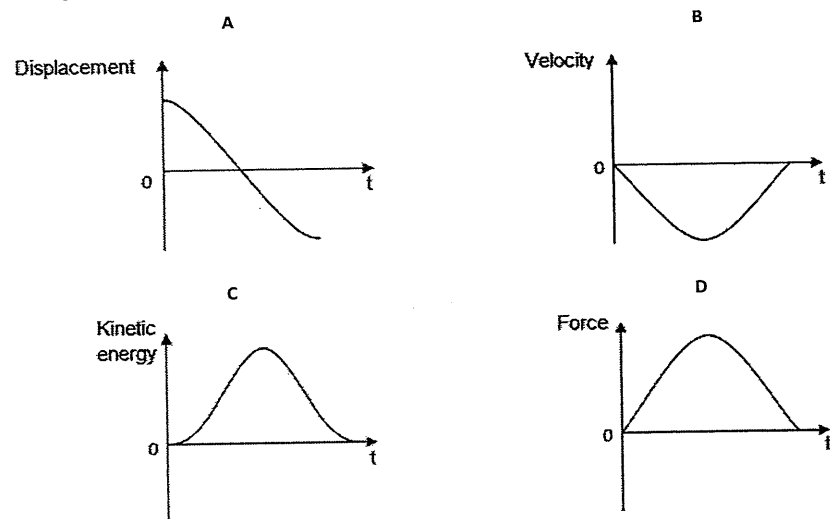
What is the period of the oscillation of the mass?

- A $\frac{2\pi}{25} \text{ s}$ B $\frac{2\pi\sqrt{2}}{5} \text{ s}$ C $\frac{8\pi}{25} \text{ s}$ D $\frac{4\pi}{5} \text{ s}$

25. [SAJC/Prelims 2014/P1/18]

A small sphere connected between two fixed supports by two identical springs is set into simple harmonic motion. The graphs below show the variation of displacement, velocity, kinetic energy and force with time t during part of the motion.

Which graph must be wrong because it does not fit with all others?



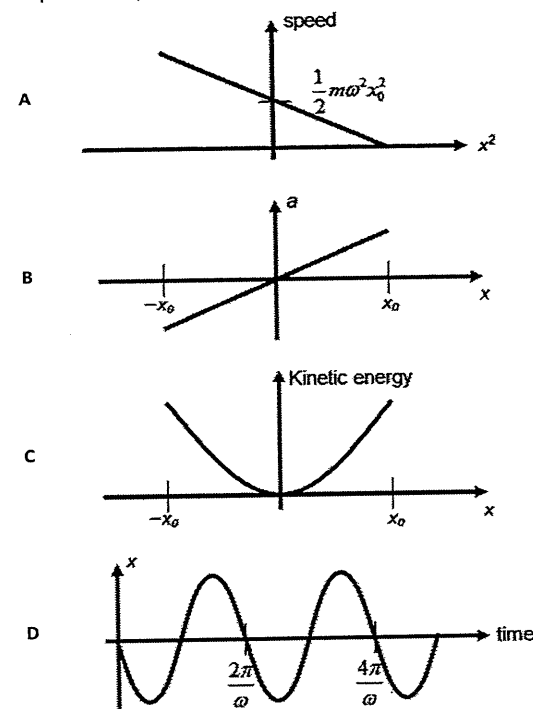
26. [SRJC/Prelims 2014/P1/15]

A simple harmonic motion of a mass has displacement from equilibrium x , amplitude x_0 and period T s. Which of the following statement is a correct description of its speed and/or acceleration at various positions?

- A After the mass passes by the equilibrium position, $T/2$ s later, it will be at displacement of $\frac{1}{2}x_0$.
- B Maximum speed and acceleration occur at equilibrium.
- C As mass moves towards equilibrium, acceleration points away from the equilibrium since acceleration is always opposite to the direction of displacement.
- D Speed at displacement of $\frac{1}{2}x_0$ is not $\frac{1}{2}$ of its maximum speed.

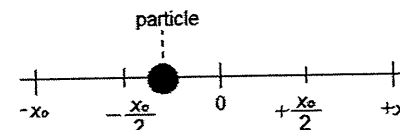
27. [SRJC/Prelims 2014/P1/16]

Which graph is the most accurate description of the simple harmonic motion of a mass m with angular velocity ω , displacement x , acceleration a and amplitude x_0 ?



28. [TJC/Prelims 2014/P1/16]

A particle of mass 0.20 kg is moving with simple harmonic motion with the amplitude $x_0 = 0.050$ m, as shown below.



If the angular frequency of the particle is 4.0 rad s^{-1} , what is the shortest time taken for it to move from $-\frac{x_0}{2}$ to $+\frac{x_0}{2}$?

- A 0.26 s
- B 0.39 s
- C 0.52 s
- D 0.65 s

29. [TJC/Prelims 2014/P1/17]

Fig. 1 shows the variation with displacement x of the velocity v of a simple harmonic oscillator.

Fig. 2 shows the variation with t of the net force F acting on the oscillator.

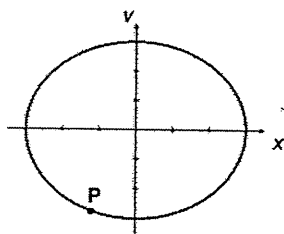


Fig. 1

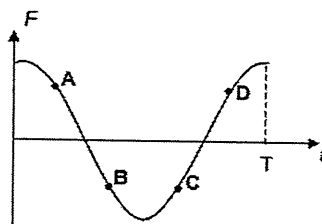
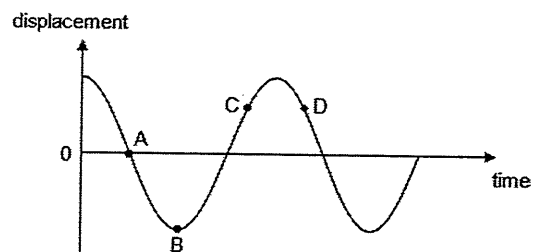


Fig. 2

Which of the points on Fig. 2 corresponds to the state of motion represented by point P on Fig. 1?

30. [TPJC/Prelims 2014/P1/14]

The diagram below shows the graph of displacement against time for a body performing simple harmonic motion. At which point are the velocity and acceleration in opposite directions?



31. [TPJC/Prelims 2014/P1/15]

A particle is oscillating in simple harmonic motion. The kinetic energy of the particle is E_k , its potential energy is E_p and the total energy is E_T . When the particle is mid-way between the equilibrium position and an amplitude position, what are the values of the ratios $E_k: E_T$ and $E_p: E_T$?

	$E_k: E_T$	$E_p: E_T$
A	1: 4	3: 4
B	1: 2	1: 2
C	2: 3	1: 3
D	3: 4	1: 4

32. [YJC/Prelims 2014/P1/15]

Which of the following is true for a damped system of oscillations?

- A The system does not oscillate at its natural frequency.
- B When resonance takes place, the system will still lose energy.
- C Damping causes the resonant frequency to become gradually bigger.
- D The number of complete oscillations made by a system under critical damping is less than the number of complete oscillations made by a system under heavy damping.

33. [YJC/Prelims 2014/P1/16]

A body performs simple harmonic motion with a period of 0.63 s. The maximum speed of the body is 5.0 m s^{-1} . What are the values of the amplitude x_0 and the angular frequency ω ?

	x_0 / m	$\omega / \text{rad s}^{-1}$
A	2.0	9.97
B	0.50	0.100
C	0.50	9.97
D	0.020	0.200

34. [YJC/Prelims 2014/P1/15]

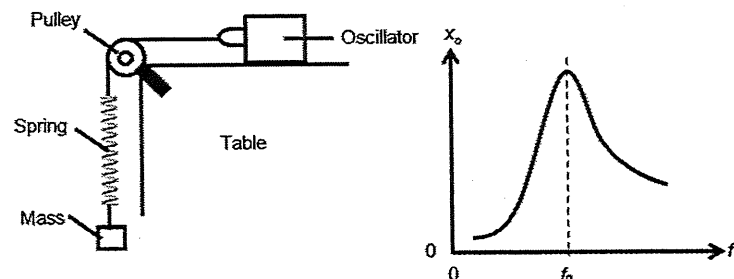
A mass of 10.0 g oscillates in simple harmonic motion with an amplitude of 5.0 mm at a frequency of 50 Hz. What is the total energy of this simple harmonic oscillator?

- A 0.0100 J B 0.0123 J C 0.0157 J D 0.0230 J

35. [YJC/Prelims 2014/P1/16]

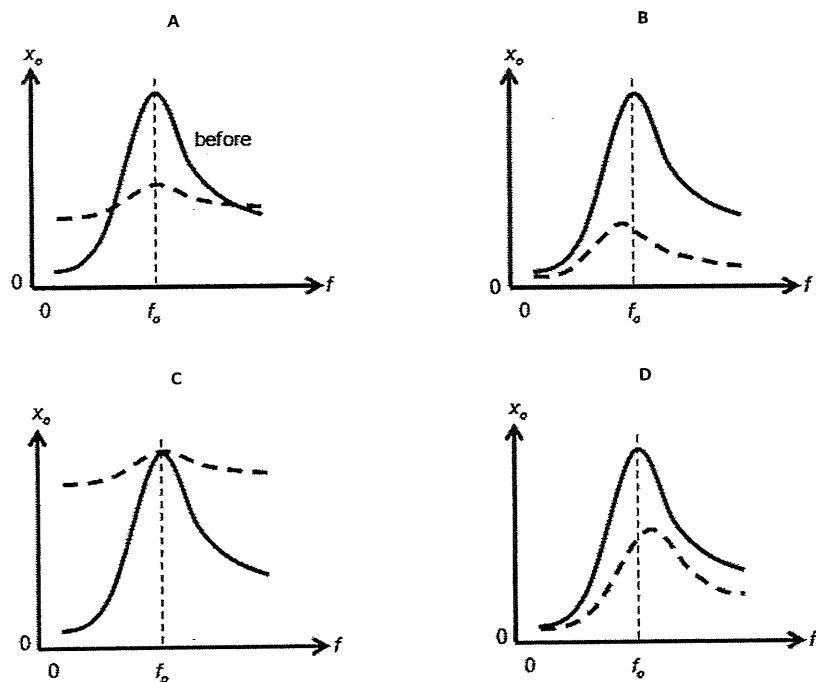
A mass, suspended from a helical spring, is made to oscillate by an oscillator.

The graph below shows the variation with the frequency f of the amplitude x_0 of the vibration of the mass.



The mass is immersed in a viscous oil solution to cause heavy damping of the oscillations.

Which graph shows how x_0 vary with f of the mass before and after being immersed in the oil solution?



II. Structured Questions

1. [AJC/Prelims 2014/P3/8]

A toy car's wheel is set up as a pendulum by hanging it vertically from a fixed support. The wheel oscillates about the fixed support as illustrated in Fig. 1.1.

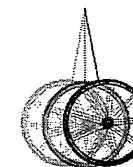


Fig. 1.1

The variation with displacement x of the acceleration a of the wheel is shown in Fig. 1.2.

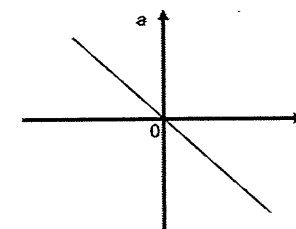


Fig. 1.2

- Use Fig. 1.2 to explain why the motion of the wheel is simple harmonic. [3]
 - State a condition that must be satisfied for the oscillation of the wheel to be simple harmonic. [1]
- The wheel in Fig. 1.1 is displaced and released at time $t = 0$. The oscillations of the wheel have amplitude 14.7 cm and angular frequency 2.40 rad s^{-1} .

 - Define angular frequency. [1]
 - State an expression for the displacement x of the wheel in terms of time t . [1]
 - Use your expression in b. ii. to sketch the variation of the kinetic energy of the wheel E_k with time t for one complete oscillation in Fig 1.3. [2]

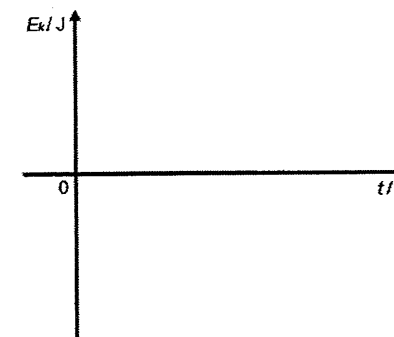


Fig. 1.3

- c. The angular frequency value stated in b. is calculated from the period of the simple harmonic motion. An accurate value for the period is found by timing a large number of oscillations.
- Explain why a large number of oscillations would help to achieve a more accurate value for the period. [1]
 - Calculate the number of oscillations that are measured in a total time of 83.8 s. [2]
- d. In order to time the oscillations of the wheel in Fig. 1.1, a stopwatch is started when the centre of the wheel passes a marker. This marker is 2.7 cm from the equilibrium position as shown in Fig. 1.4.

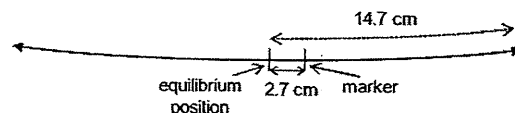


Fig. 1.4

- The wheel has a mass of 0.165 kg.
Calculate the restoring force acting on the wheel as it passes the marker. [2]
 - Calculate the speed of the wheel at the instant it passes the marker. [2]
- d. In an experiment to demonstrate resonance, the wheel in Fig. 1.1 is made to oscillate by an external periodic driving force of frequency f . Fig. 1.5 shows the variation with frequency f of the amplitude of the forced oscillations of the wheel.

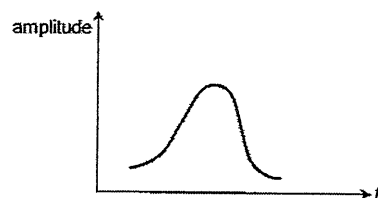


Fig. 1.5

- Describe how the amplitude of oscillation depends on the frequency of the driving force. [2]
- On Fig. 1.5, sketch the shape of the graph if the forced oscillations of the wheel were repeated in a vacuum. Label your sketch as A. [2]

[Total: 19]

2. [DHS/Prelims 2014/P3/1]

The arrangement shown in Fig. 2.1 is used to determine the length x of a spring when different masses M are attached to the spring.

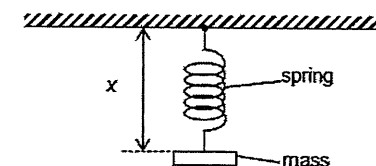


Fig. 2.1

The variation with length x of mass M is shown in Fig. 1.2.

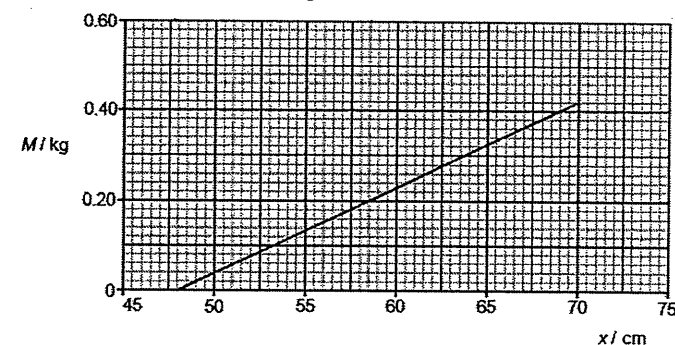


Fig. 2.2

- State the natural length of the spring. [1]
- Use Fig. 2.2 to determine the spring constant of the spring. [2]
- A mass m is then attached to the spring. The mass is displaced vertically then released. The variation with time t of the displacement y from its mean position is shown in Fig. 2.3.

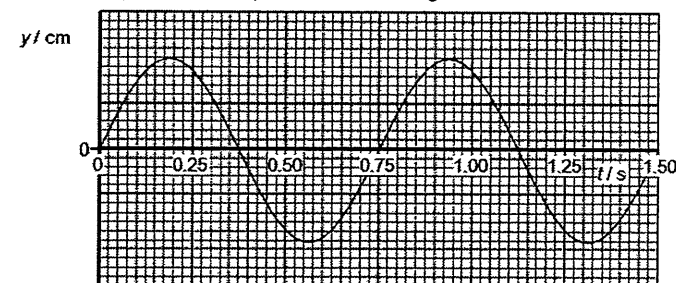


Fig. 2.3

Calculate the angular frequency of the oscillation.

[1]

d. The spring-mass system undergoes free oscillations.

- Explain what is meant by free oscillations. [1]
- Determine the mass m that results in the oscillations shown in Fig. 2.3. [2]
- Hence or otherwise, find the length x of the spring at $y = 0$ cm. [1]

[Total: 8]

3. [HCI/Prelims 2014/P2/2]

Fig. 3.1 shows a device for measuring the frequency of vibrations of an engine. The rigid metal probe and case is held against the engine so that the engine vibrations induce forced vibrations of the strip in the direction shown.

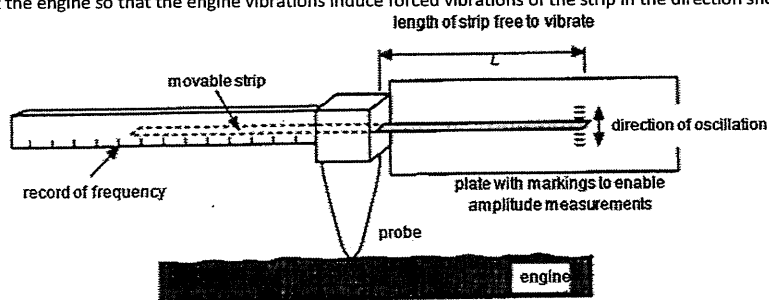


Fig. 3.1

The length L of the strip which is free to vibrate can be varied until a maximum amplitude is observed. The frequency corresponding to L can then be read from the scale.

- State what will be observed as the length L of the strip that is free to move is gradually increased from a small value. [1]
 - Explain why this happens. [2]
- In one test, the end of the strip oscillated with simple harmonic motion of amplitude 5.0 mm at a frequency of 12 Hz.
 - Define simple harmonic motion. [2]
 - Determine for the end of the strip,
 - its maximum speed [2]
 - its maximum acceleration [1]

- c. The free end of the vibrating strip is an antinode and the end near the probe is a node. When the natural frequency of the strip is 12 Hz, the length L of the strip that is free to vibrate is 16 cm. This has been plotted for you in Fig. 3.2.

Sketch a graph on Fig. 3.2, showing how the length L varies with natural frequency of vibration of the strip. [2]

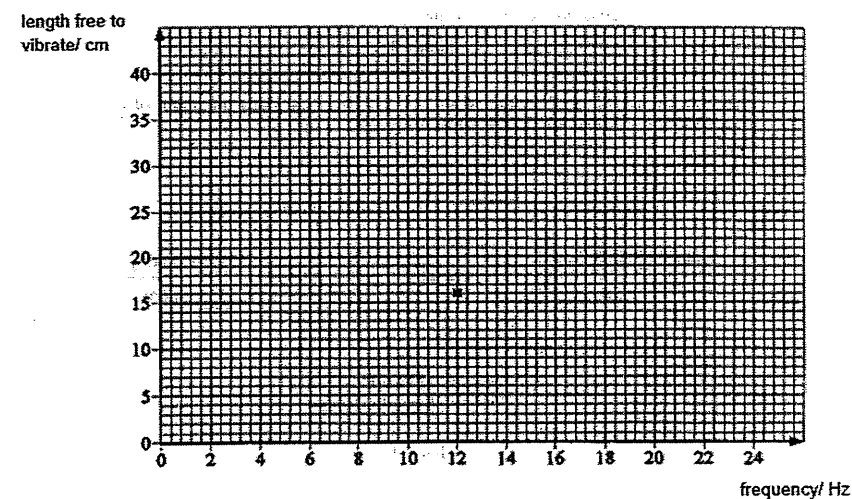


Fig. 3.2

[Total: 10]

4. [JC/Prelims 2014/P3/1]

Fig. 4.1 shows a diagram of a pendulum that was raised through a height of 2.4 cm from its lowest position P to a position Q.

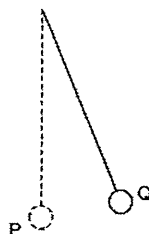


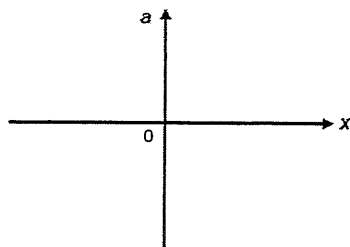
Fig. 4.1

- Indicate the forces acting on the pendulum at Q with clearly-labelled arrows. [2]
- A body can be considered to be in equilibrium at times. State the conditions for a body to be in equilibrium. [2]
- The pendulum is released from Q such that it oscillates about P with a period of 1.3 s.
 - Explain qualitatively what happens to the tension in the string of the pendulum when it first passes P. [2]
 - Calculate the speed of the pendulum bob when it first passes P. [2]
 - Using your answer to ii., determine the amplitude of oscillation of the pendulum bob about P. [2]

[Total: 8]

5. [JJC/Prelims 2014/P3/8(a)]

- Some sand is placed on a flat horizontal plate and the plate is made to oscillate with simple harmonic motion in a vertical direction, with amplitude x_0 and a frequency of 13 Hz.
 - Sketch a graph to show the variation with displacement x of the acceleration a of the plate. [2]



- Calculate the amplitude of oscillation of the plate such that the maximum acceleration is numerically equal to the acceleration of free fall. [2]
- Suggest, with a reason, what happens to the sand on the plate when the amplitude of oscillation of the plate exceeds the value calculated in ii. [3]

[Total: 7]

6. [NJC/Prelims 2014/P3/4]

- Define simple harmonic motion.
Explain how you would determine whether such a motion remains simple harmonic at large amplitude. [3]
- A 1500 kg car has a suspension system that sags 5.0 cm when the chassis is placed on it. The energy of the oscillations, E , is given by

$$E = \frac{1}{2} k x_m^2 e^{-\frac{bt}{m}}$$

where x_m is the amplitude of the oscillations at $t = 0$ s

m is the mass of the system

k is the spring constant

b is the damping constant of 19800 kg s^{-1}

- Calculate the spring constant, k . [1]
 - Determine the time required for the amplitude of oscillations to fall to one-third of its initial value. [3]
- A 207 kg BMW motorcycle carrying its driver and pillion passenger each at 80 kg travels over a "washboard" dirt road with corrugations 2.0 m apart. The motorcycle bounces with maximum amplitude when its speed is 20 km h^{-1} .
When the motorcycle is parked, and the driver and pillion passenger got off, calculate how much the motorcycle rises on its suspension using the expression

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \quad [3]$$

[Total: 10]

7. [NYJC/Prelims 2014/P3/6]

- A structure consists of a sphere of mass 0.500 kg, attached firmly to one end of a light rod. The other end of the rod is pivoted freely at point O. The distance between the centre of gravity of the sphere to O is 0.400 m. The structure is held in a position such that the rod is at an angle of 5.00° from the vertical, as shown in Fig. 7.1. The structure is then released from rest and oscillates in simple harmonic motion. At one instant during the oscillation, the sphere is directly below O, as shown in Fig. 7.2.

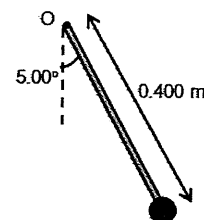


Fig. 7.1

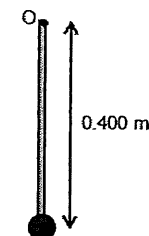


Fig. 7.2

- On Fig. 7.2, indicate the forces acting on the structure. [2]
- Show that the linear speed of the sphere at the instant in Fig. 7.2 is 0.173 m s^{-1} . [2]

- iii. Determine the force exerted on the pivot by the structure in Fig. 7.2 [3]
- iv. With the aid of a diagram, discuss whether the force exerted by the pivot on the structure is always upward throughout the oscillation. [3]
- b. A horizontal turntable is connected to a motor such that it rotates at exactly 47 revolutions per minute. A peg is fixed vertically on the turntable. A horizontal beam of light casts a shadow of the peg onto a screen in front of which is suspended the structure mentioned in a., as shown in Fig. 6.3.

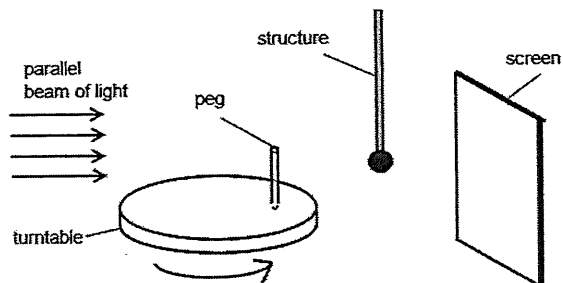


Fig. 7.3

The length of the structure is adjusted such that the shadows of the peg and the structure move in phase on the screen.

- i. Explain what is meant by in phase in this context. [1]

The angular speed of the turntable suddenly increases to $47\frac{1}{3}$ revolutions per minute.

- ii. Define angular speed. [1]
- iii. Briefly describe what will be observed on the screen. [2]
- iv. Determine the time taken before the two shadows are next in phase. [2]
- v. Calculate the number of oscillations made by the peg before the shadows are next in phase. [2]
- vi. State two assumptions that were made in the calculations of b. iv. and b. v. [2]

[Total: 20]

8. [PJC/Prelims 2014/P3/2]

- a. A partially submerged wooden block is displaced vertically and then released. For a displacement x that is measured from its equilibrium position, the acceleration a of the wooden block is given by the expression

$$a = -\frac{28}{m}x$$

where m is the mass of the wooden block.

- i. Explain why the expression leads to the conclusion that the block is performing simple harmonic motion. [3]
- ii. Fig. 8.1 shows the variation of kinetic energy of the block with time.

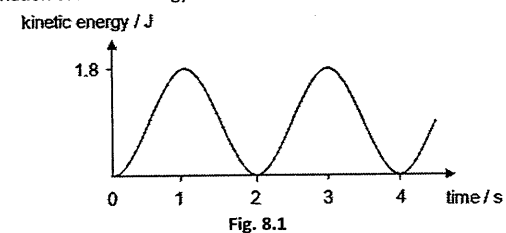


Fig. 8.1

Calculate

1. the frequency for the oscillations of the block [1]
2. the mass of the block [2]
3. the amplitude of the oscillations. [2]

- b. On Fig. 8.2, sketch a labelled graph of the variation of the displacement with time of the wooden block. [2]

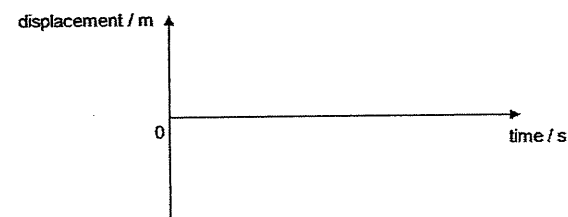


Fig. 8.2

[Total: 10]

9. [VHS/Prelims 2014/P3/2]

- a. State what is meant by resonance. [2]
- b. A sailing boat of 1020 kg undergoes simple harmonic motion in the vertical direction due to surface water waves that are incident on the boat as shown in Fig. 9.1. The water waves of amplitude of 0.18 m and frequency 0.56 Hz cause resonance in the vertical motion of the boat.

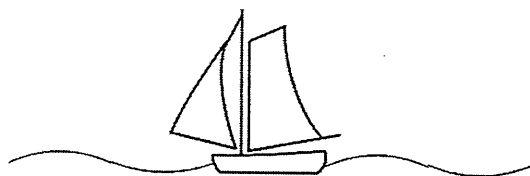
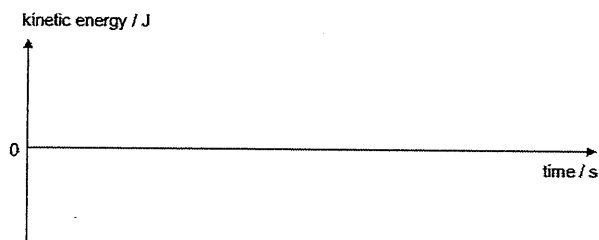


Fig. 9.1

- i. Derive an expression for the vertical displacement x as a function of the time t associated with the motion of the boat. [2]
- ii. Calculate the magnitude of acceleration of the sailing boat when its displacement from its equilibrium position is 0.080 m. [2]
- iii. Calculate the maximum kinetic energy of the sailing boat during the vertical oscillation. [2]
- iv. Sketch on the axes below to show how the kinetic energy of the sailing boat varies with time. [2]



[Total: 10]

10. [SAJC/Prelims 2014/P3/6(a),(b)]

- a. A car engine is also known as an internal combustion engine, and is designed to use small, controlled explosions to create the power necessary to move the vehicle. All car engines are designed to use a four stroke combustion cycle. The four strokes are the intake, compression, combustion and exhaust. These strokes are repeated in quick succession to generate power. The core of the car engine is the cylinder, with the piston moving up and down inside the cylinder. A small, controlled explosion forces the piston of the engine to move. The piston is attached to a constantly turning axle called a crankshaft. The crankshaft powers the car's gearbox which, in turn, drives the wheels. When the energy from the explosion is almost worn out, another explosion occurs, forcing the pistons to move again. This recurring cycle generates the power needed.

A cylinder and piston, used in a car engine, are illustrated in Fig. 10.1.

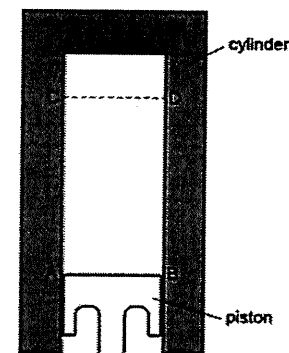


Fig. 10.1

The vertical motion of the piston in the cylinder is assumed to be simple harmonic. The top surface of the piston is at AB when it is at its lowest position; it is at CD when at its highest position, as marked in Fig. 10.1.

The displacement d of the piston may be represented by the equation

$$d = -4.0 \cos(220t)$$

where d is measured in centimetres.

- i. State the distance between the lowest position AB and the highest position CD of the top surface of the piston. [1]
- ii. Determine the number of oscillations made per second by the piston. [2]
- iii. On Fig 10.1, draw a line to represent the top surface of the piston in the position where the speed of the piston is maximum. [1]
- iv. Calculate the maximum speed of the piston. [1]

- b. The engine of a car has several cylinders. Two of these cylinders are shown in Fig. 10.2.

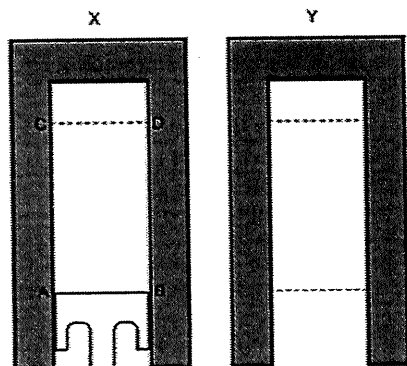


Fig. 10.2

X is the same cylinder and piston as in Fig. 10.1. Y is another cylinder, with the lowest and highest positions of the top surface of each piston indicated. The pistons in the cylinders each have the same frequency of oscillation, but they are not in phase.

At a particular instant in time, the position of the top of the piston in cylinder X is as shown.

In cylinder Y, the oscillations of the piston lead those of the piston in cylinder X by a phase angle of 120° ($\frac{2}{3}\pi$ rad).

- Show that the displacement of the piston in Y for this instant is 2.0 cm. [1]
- Calculate the speed of the piston in Y for this instant. [1]
- Complete the diagram of cylinder Y, for this instant, by drawing
 - a line to show the top surface of the piston [1]
 - an arrow to show the direction of movement of the piston. [1]

[Total: 9]

11. [TJC/Prelims 2014/P3/2]

Fig. 11.1 shows a dog standing in the middle of a poorly designed bridge which is oscillating up and down with simple harmonic motion with a natural period of 0.80 s and an amplitude of 10.4 cm.

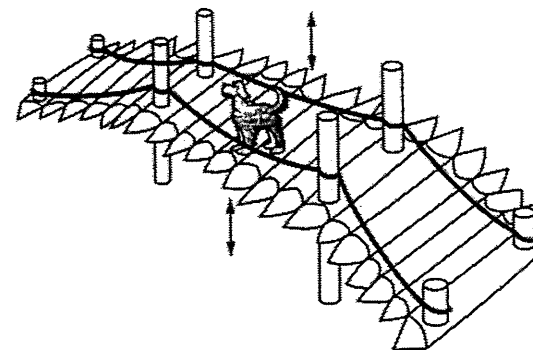


Fig. 11.1

- State what is meant by simple harmonic motion. [1]
- Calculate the maximum speed of the oscillating bridge. [2]
- The amplitude of oscillation of the bridge is increased. The variation of the displacement of the bridge with time is shown in Fig. 11.2. At $t = 0$ s, the dog is at the equilibrium position of the bridge and the bridge is moving downwards.

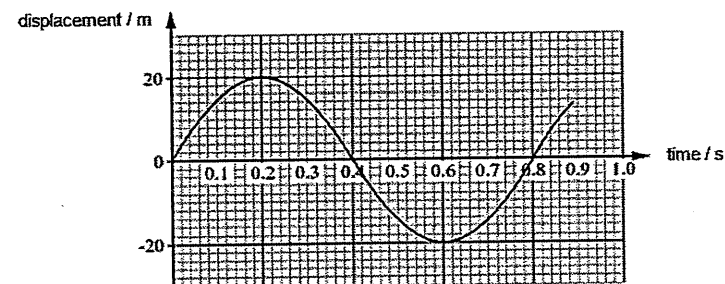


Fig. 11.2

- Calculate the displacement of the bridge when the dog first loses contact with the bridge, and determine from the graph the time at which this happens. [4]
- Without further calculations, label with the letter A on Fig. 11.2 the point at which the contact force acting on the dog is the greatest. [1]

[Total: 8]

12. [TPJC/Prelims 2014/P3/2]

- a. Describe how, for a simple harmonic motion, the direction of acceleration varies with the direction of the velocity. [2]
- b. A spring with spring constant k is loaded with a mass m and extends by a length L as shown in Fig. 12.1.

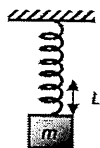


Fig. 12.1

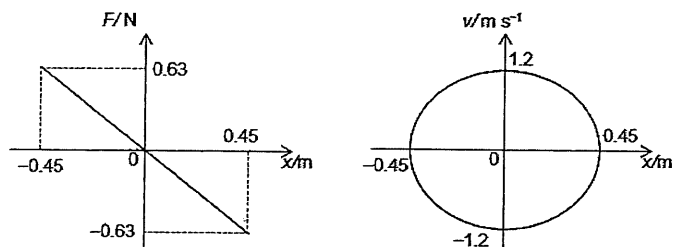
Using Hooke's Law, write down an expression for the tension acting on the mass. [1]

- b. By considering the resultant force on the mass when it is displaced downwards by a small distance x from the equilibrium position,
- draw the free body diagram of the mass at the instant of displacement x . [1]
 - show, using Newton's laws that the acceleration a of mass m can be expressed as $a = -\left(\frac{k}{m}\right)x$. [2]
 - explain why the expression in c. ii. leads to the conclusion that the mass is performing simple harmonic motion. [2]

[Total: 8]

13. [VIC/Prelims 2014/P2/1]

A mass of mass m was suspended from the bottom of a vertical spring and set into oscillation with a period T . Measurements were made, and the resultant force F acting on the mass, and its velocity v were obtained. The graphs below were then plotted to show how F and v vary with the displacement x of the mass from its equilibrium position.



Calculate

- the period of the oscillation [2]
- the mass m . [2]

[Total: 4]

14. [YJC/Prelims 2014/P3/3]

A box is placed on top of the plate which is supported by a spring as shown in Fig. 14.1. The mass of the box is 1.0 kg and the mass of the plate is 0.50 kg.

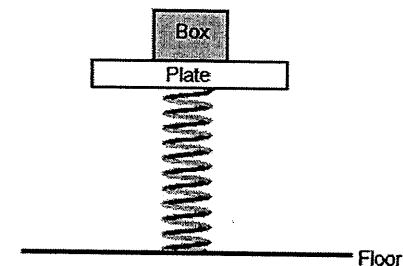


Fig. 14.1

- Both the box and plate are in equilibrium. The spring constant is 150 N m^{-1} . Calculate the compression of the spring. [2]
- The box and the plate are pushed further downwards and released. As result, the box and the plate oscillate up and down in simple harmonic motion. Calculate the maximum distance that the box can be pushed below its equilibrium position such that the box remains in contact with the plate as it oscillates. [3]
- Fig. 14.2 shows the kinetic energy of the box with its displacement from its equilibrium position.

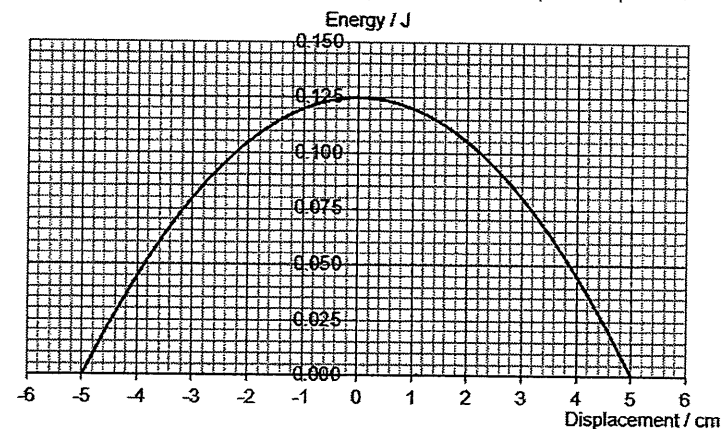


Fig. 14.2

- Calculate the period of the box. [2]
- Calculate the total energy of the box at a displacement of -4.00 cm , assuming that gravitational potential energy is zero at the lowest point of oscillation where the displacement is -5.00 cm . [1]

[Total: 8]

III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
C	C	B	B	D	C	D	C	D	B	A	B	A	C	D

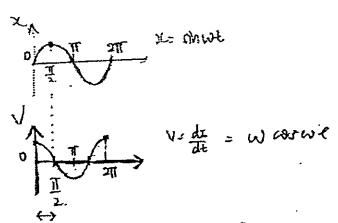
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
D	C	B	A	B	B	C	D	B	D	D	D	A	D	C

31	32	33	34	35
D	B	C	B	B

1 C Definition of SHM

2 C Since $\omega = \frac{2\pi}{T}$, when period is halved, the new angular frequency $\omega' = 2\omega$.
From $a = \omega^2 x$, the gradient of the new line will be 4 times that of the original.

3 B



$x = A \sin \omega t$
 $v = \frac{dx}{dt} = A \omega \cos \omega t$

4 B For no damping, the maximum amplitude is infinity.

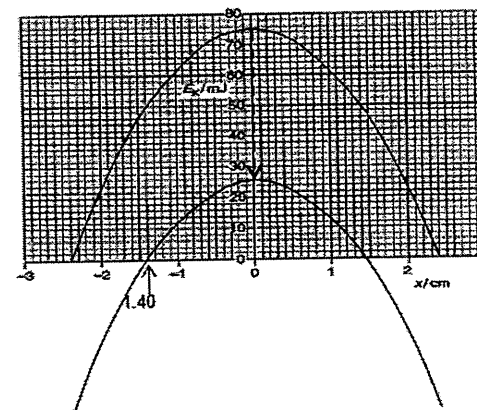
5 D angular frequency, $\omega = 2\pi f = 2\pi \left(\frac{3}{0.76 \times 10^{-3}} \right) = 24800 \text{ rad s}^{-1}$

6 C First method

Shift the graph down by 50 mJ.

Read off the x-intercept value.

$x_0 = 1.40 \text{ cm}$



Second method

$$KE_{\max} = \frac{1}{2} m \omega^2 x_0^2$$

$$KE_{\max} \propto x_0^2$$

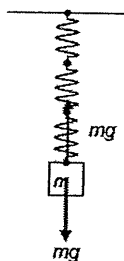
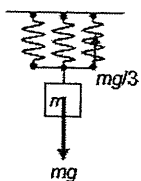
$$\frac{75}{75 - 50} = \left(\frac{2.4}{x_0} \right)^2$$

$$x_0 = 1.39 \text{ cm}$$

7 D With light damping, there should be a progressive decrease in the amplitude of oscillation.
Option A is wrong because the first two peaks are of the same height.
Option B is wrong because the amplitude is not continuously decreasing.
Option C is wrong because the graph is not sinusoidal.
Option D is true as the amplitude decreases progressively.

8 C Options A and B are wrong because based on the equation, $KE_{\max} = \frac{1}{2} m \omega^2 x_0^2$ the maximum KE for both horizontal and vertical systems are equal since k is the same, mass of P and Q are equal and the amplitude is the same.
Option C is true since for block P, the spring is not stretched at equilibrium before it is allowed to oscillate whereas for block Q, the spring is stretched at equilibrium before it oscillates with the same amplitude as block P. Hence $EPE_{\max}$ for block P < $EPE_{\max}$ for block Q.
Based on the reasoning for option C, option D is wrong.

9 D At D, the velocity is +ve, but is decreasing. This means that the particle is moving away from the equilibrium point, towards +x₀ (to the right). But acceleration is towards x = 0 (to the left), so v and a are opposite at D.



Revision Package (H2 Physics)

9 – Oscillations

Tension in each spring is $mg/3$.

Let the extension in each spring be x .

By Hooke's Law, $mg/3 = kx$.

$$mg = 3kx.$$

But for the system as a whole, $mg = k_{\text{equiv}} x$.

Comparing the two expressions (in mg), we get $k_{\text{equiv}} = 3k$.

In this set-up of springs in series, the tension in each spring is mg .

Let each spring be extended by x .

Then, $mg = kx$.

The total extension for the system as a whole is $3x$.

Therefore, $mg = k_{\text{equiv}} (3x) = (3k_{\text{equiv}}) x$.

Comparing the two equations (in mg), we get,

$$3k_{\text{equiv}} = k,$$

$$k_{\text{equiv}} = k/3$$

Since $\omega = 2\pi f = \sqrt{\frac{k}{m}}$, option B has largest k , so it has the highest frequency.

- 11 A $d = 0.25 - 0.03 \cos 2.5\pi t$ means that amplitude $A = 0.03$ m, and $\omega = 2.5\pi$

$$\omega = \frac{2\pi}{T} \Rightarrow T = \frac{2\pi}{2.5\pi} = 0.80 \text{ s}$$

$$\text{Max KE} = \frac{1}{2} m \omega^2 A^2 = \frac{1}{2} \times 0.300 (2.5\pi)^2 (0.03)^2 = 8.3 \text{ mJ}$$

Remember that 2 cycles on the KE against time graph is 1 cycle on the displacement against time graph.

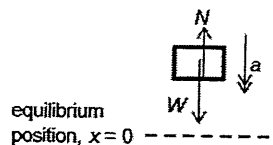
- 12 B Consider the free-body diagram of the object:

$$mg - N = ma$$

If the body loses contact with the platform, $N = 0$,

$$\therefore mg = ma \Rightarrow g = a \Rightarrow g = \omega^2 x_0$$

$$x_0 = \frac{g}{\omega^2} = \frac{9.81}{(2\pi \times 13)^2} = 1.5 \text{ mm}$$



- 13 A Object has maximum PE at $t = 0$ means that object starts at either extreme position at $t = 0$.
Since acceleration is proportional to displacement, the acceleration should be either a positive maximum or negative maximum at $t = 0$.
- 14 C At critical damping, the object will return to its equilibrium position in the shortest possible time.

- 15 D By attaching a heavy mass, the natural frequency of the drum will be lowered. Hence resonance will now occur at a lower driving frequency. Note that attaching a heavy mass does not change the damping.

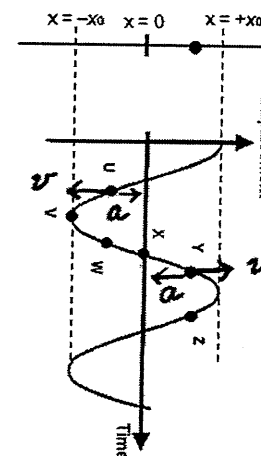
- 16 D The value of the potential energy increases as the body moves away from point X suggests that the force is directed towards X. Hence the body will oscillate about point X.

Note that force $F = -\frac{dE_p}{dx}$. E.g. on the RHS of curve, gradient is positive, hence F will be negative (since $F = -\frac{dE_p}{dx}$), hence the force is directed towards the left (i.e. to the centre).

- 17 C In SHM, $\omega^2 = \frac{k}{m}$. For spring-mass systems, period $T = 2\pi\sqrt{\frac{m}{k}}$
But, for simple pendulums, it can be shown that $T = 2\pi\sqrt{\frac{L}{g}}$, that is, T is independent of mass of body.

- 18 B Speed $v = \omega\sqrt{A^2 - x^2} = 2\pi(2.0)\sqrt{2.0^2 - 1.0^2} = 22 \text{ cm s}^{-1}$

- 19 A If we imagine that the particle is undergoing SHM along a horizontal line (see diagram below), then, by rotating the graph by 90° , it becomes very clear that at positions Y and U, the particle is travelling and accelerating in opposite directions.



- 20 B The minimum force occurs when the piston is accelerating downwards with maximum acceleration.

$$a_{\text{max}} = \left| -\omega^2 x_0 \right| = \left(\frac{2\pi}{0.5} \right)^2 (0.02) = 3.2 \text{ m s}^{-2}$$

$$\text{Minimum force} = m(g - a_{\text{max}}) = 2.0 \text{ N}$$

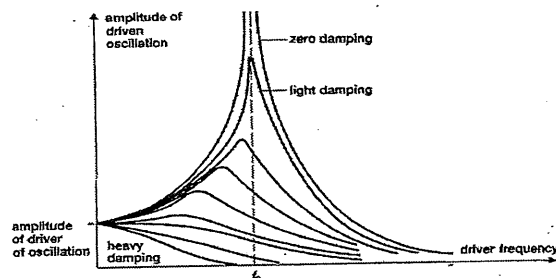
21 B $a = -\omega^2 x_0 \Rightarrow 6 = \omega^2 (0.3) \Rightarrow \omega = 4.5 \text{ rad s}^{-1}$

$v = \omega x_0 = (4.5)(0.3) = 1.3 \text{ m s}^{-1}$

22 C $v = \omega \sqrt{x_0^2 - x^2} = (2\pi(1.0)) \sqrt{4^2 - 3^2} = 16.6 \text{ cm s}^{-1}$

- 23 D When damping increases, resonant frequency decreases.

Since $f = \frac{1}{T}$, the period of forced oscillation will increase.



24 B $PE_{\max} = KE_{\max} = \frac{1}{2} m \omega^2 A^2$

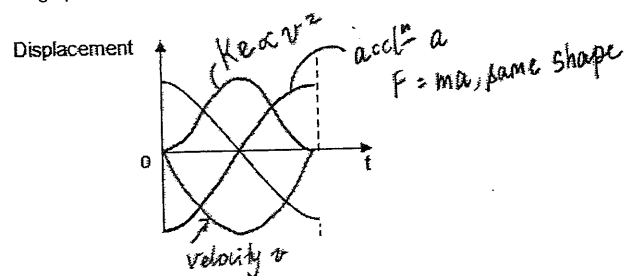
$$\omega = \frac{1}{A} \sqrt{\frac{2PE_{\max}}{m}} = \frac{1}{0.2} \sqrt{\frac{2 \times 1.0}{4}} = \frac{5}{\sqrt{2}}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi\sqrt{2}}{5}$$

- 25 D Let's sketch in the velocity-time, KE-time and Force-time graphs on the displacement-time graph and then compare with the options to pick the right answer.
In sketching the graphs, just remember:

$v = \frac{dx}{dt}$ = gradient of $x-t$ graph, $a \propto -x$ for SHM

$F = ma$, so $F-t$ graph has the same shape as the $a-t$ graph



- 26 D A: To move from equilibrium to half amplitude, time is $T/8$ s.

B: Acceleration at equilibrium is zero.

C: Displacement is a position, not a movement. Acceleration always points towards equilibrium.

D: At $\frac{1}{2} x_0$ is speed is more than $\frac{1}{2}$ of its maximum speed

- 27 D A is wrong because x^2 cannot be negative.

B is wrong because a and x must have opposite signs.

C is wrong because KE is max at equilibrium position and min at amplitude.

- 28 A $\omega = 4.0 \text{ rad s}^{-1}$

Taking the timing to start at the equilibrium position,

$$x = x_0 \sin \omega t$$

$$\frac{x_0}{2} = x_0 \sin 4.0t$$

$$t = \frac{1}{4} \sin^{-1} 0.50 = 0.13 \text{ s}$$

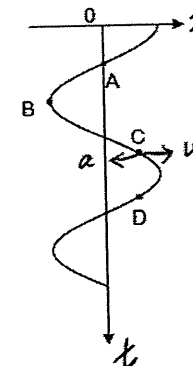
Hence total time = $0.13 \times 2 = 0.26 \text{ s}$

- 29 D At point P, the displacement is negative and velocity is negative.

Since $a = -\omega^2 x \Rightarrow F = ma = -m\omega^2 x$, x negative will mean F is positive (B and C no longer plausible).

Since $a-t$ graph is a cosine graph, $v-t$ graph will be a sine graph and hence A is not possible.

- 30 C Point C is moving towards the $+x$ direction (as it hasn't reached the amplitude point yet). But the acceleration at C is towards the equilibrium point, i.e., in the $-x$ direction.
[Alternatively, like before in Q20, consider a particle moving in SHM along a horizontal line. Turn the graph by 90° to see more quickly].



31 D $E_{\text{total}} = \frac{1}{2}m\omega^2 A^2$, $E_p = \frac{1}{2}m\omega^2 x^2$

$$\frac{E_p}{E_{\text{total}}} = \frac{\frac{1}{2}m\omega^2 x^2}{\frac{1}{2}m\omega^2 A^2} = \left(\frac{x}{A}\right)^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4} \Rightarrow \therefore \frac{E_k}{E_{\text{total}}} = \frac{3}{4}$$

32 B Damping means that energy is lost from a system over time, regardless of whether that system is in a state of resonance or not.

33 C $\omega = \frac{2\pi}{T} = \frac{2\pi}{0.63} = 9.97 \text{ rad s}^{-1}$

$$v_0 = \omega x_0$$

$$x_0 = \frac{5.0}{9.97} = 0.50 \text{ m}$$

34 B $E_{\text{total}} = \frac{1}{2}m\omega^2 A^2 = \frac{1}{2} \times 0.0100 \times (2\pi \times 50)^2 \times (5.0 \times 10^{-3})^2 = 0.0123 \text{ J}$

35 B The oil increases the damping on the oscillator. When this happens, the resonance peak falls and the resonance frequency decreases.

Structured Questions

1. [AJC/Prelims 2014/P3/8]

- a. i. SHM is the motion of a body such that its acceleration is proportional to its displacement from a fixed point and is always directed towards that point.

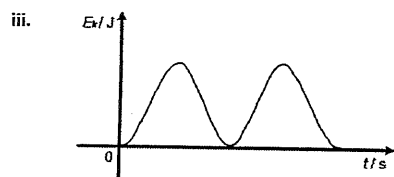
The oscillations are simple harmonic because:

The straight line passes through the origin indicating that the acceleration a of the plate is directly proportional to the displacement x from its equilibrium position

The negative gradient of the graph indicates that its acceleration a is opposite in direction to its displacement x , and is directed towards the equilibrium position.

Note: acceleration and displacement not inversely proportional

- ii. The angle of oscillation must be small.
- b. i. Angular frequency ω of a body undergoing simple harmonic motion is a constant of a given oscillator and is related to its natural frequency f by $\omega = 2\pi f$.
- ii. $x = 0.147 \cos(2.40t)$ OR $x = -0.147 \cos(2.40t)$
OR $x = 14.7 \cos(2.40t)$ OR $x = -14.7 \cos(2.40t)$



- e. i. To reduce fractional error of time measurement.

ii. $T = \frac{2\pi}{\omega} = \frac{2\pi}{2.40} = 2.6180 \text{ s}$

$$\text{number of oscillations} = \frac{83.8}{2.6180} = 32 \text{ (round to lower integer)}$$

d. i. $F = ma = m\omega^2 x = (0.165)(2.40)^2 (0.0270) = 0.0257 \text{ N}$

ii. $v = \omega \sqrt{x_0^2 - x^2} = (2.40) \sqrt{(0.147)^2 - (0.0270)^2} = 0.347 \text{ m s}^{-1}$

OR

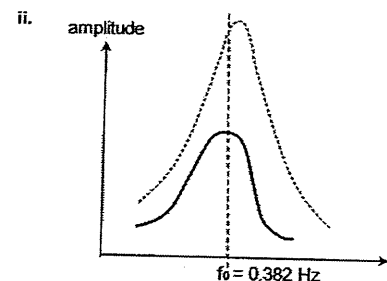
$$x = x_0 \cos \omega t$$

$$2.7 = 14.7 \cos(2.40t)$$

$$t = 0.57753 \text{ s}$$

$$v = \omega x_0 \sin \omega t = 2.40(14.7 \times 10^{-2}) \sin(2.40 \times 0.57753) = 0.347 \text{ m s}^{-1}$$

- e. i. The amplitude of oscillation increases when f increases to natural frequency f_0 . When f is larger than f_0 , the amplitude decreases. Amplitude is maximum when the frequency is equalled to f_0 .
(Learning point: $f_0 = 1/T$ or $\omega/2\pi = 2.40/2\pi = 0.382 \text{ Hz}$)



2. [DHS/Prelims 2014/P3/1]

- a. 0.4800 m (half smallest square $\rightarrow$ 2 d.p.)

b. $k = \frac{\text{force}}{\text{extension}} = \frac{mg}{x - l_{\text{natural}}} = \frac{(0.42 - 0.00)(9.81)}{(0.7000 - 0.4800)} = 18.7 \text{ N m}^{-1}$

c. $\omega = \frac{2\pi}{T} = \frac{2\pi}{0.75} = 8.38 \text{ rad s}^{-1}$

- d. i. oscillations in the absence of an external force / with no loss or gain in energy / constant amplitude

ii. $-ky = ma$

$$a = -\frac{k}{m}y$$

$$\omega^2 = \frac{k}{m}$$

$$m = \frac{k}{\omega^2} = \frac{18.7}{8.38^2} = 0.266 \text{ kg}$$

$$\text{iii. } mg = kx_o \Rightarrow x_o = \frac{(0.266)(9.81)}{18.7} = 14.0 \text{ cm}$$

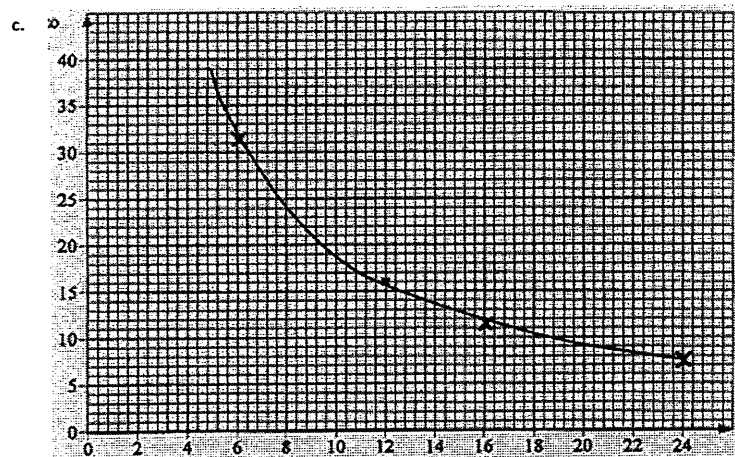
$$x = x_o + l_{\text{natural}} = 0.140 + 0.4800 = 0.620 \text{ m}$$

OR

$$\text{read from } m_1 = 0.26 \text{ (nearest half square)} \Rightarrow x = 0.615 \text{ to } 0.625 \text{ m}$$

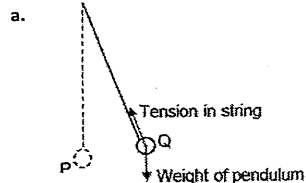
3. [HCI/Prelims 2014/P2/2]

- a. i. The amplitude increases to a maximum and then decreases.
 ii. The strip has a natural frequency which depends on its length. Maximum amplitude when the frequency of engine vibration equals to the natural frequency of the length of strip free to vibrate.
- b. i. The magnitude of the acceleration is proportional to the displacement from its equilibrium position and is directed opposite to the displacement/ directed towards the equilibrium position.
- maximum speed = $\omega x_o = 2\pi f x_o = 2\pi(12)(0.005) = 0.38 \text{ m s}^{-1}$
 - maximum acceleration = $\omega^2 x_o = 2\pi f x_o = 4\pi^2(12)^2(0.005) = 28 \text{ m s}^{-2}$



The length L is equivalent to $\frac{1}{4}$ wavelength $\Rightarrow L = \frac{v}{4f} \Rightarrow Lf = \text{constant} = 16 \times 12$

4. [JJC/Prelims 2014/P3/1]

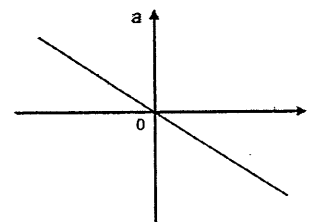


- b. See definitions in lecture notes.

- c. i. The tension is larger than the weight of the pendulum because a net/resultant force towards the centre of the circle is required for the bob to execute circular motion.
- ii. The decrease in GPE = increase in KE
 $mg(0.024) = \frac{1}{2}mv^2$ (since initial KE = 0)
 $v = 0.686 \text{ m s}^{-1}$
- iii. When the pendulum bob passes P, its speed is a maximum (oscillating object at equilibrium position).
 $v_o = \omega x_o \Rightarrow 0.686 = (2\pi/1.3)(x_o) \Rightarrow x_o = 0.142 \text{ m}$

5. [JJC/Prelims 2014/P3/8(a)]

- a. i.



ii. Numerically, $a_o = \omega^2 x_o \Rightarrow \text{amplitude, } x_o = \frac{a_o}{\omega^2} = \frac{a_o}{(2\pi f)^2} = \frac{9.81}{[2\pi(13)]^2} = 1.47 \times 10^{-3} \text{ m}$

- iii. The sand will lose contact with the plate for some time in the oscillation. When the amplitude exceeds that in ii., the acceleration of the plate also exceeds the acceleration of free fall for some time in the oscillation. In that duration, the sand accelerating with the acceleration of free fall cannot catch up with the motion of the plate, hence it will lose contact with the plate.

6. [JJC/Prelims 2014/P3/4]

- a. See definitions in lecture notes.

Compare the period of the oscillations at large amplitude to the period of the simple harmonic oscillations at small amplitude. If the periods are the same, the oscillation at large amplitude is still in simple harmonic. If the periods are different, the oscillation at large amplitude is no longer simple harmonic.

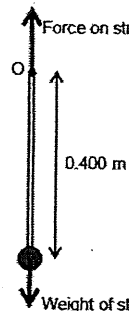
- b. i. $F = kx$
 $1500(9.81) = k(0.05)$
 $k = 2.94 \times 10^5 \text{ N m}^{-1}$

- ii. $E(0) = 0.5 kx_m^2 \text{ --- (1)}$
 $E(t) = 0.5 kx_m^2 e^{-bt/m} \text{ --- (2)}$
 When amplitude decreases by 1/3, Energy decreases by 1/9
 $1/9 = e^{-19800t/1500}$
 $1500 \times \ln(9)/19800 = t$
 $t = 0.166 \text{ s}$

- c. $\lambda = 2.0 \text{ m}$
 $v = 20 \text{ km/h} = 5.56 \text{ m s}^{-1}$
driving frequency $= v / \lambda = 5.56/2 = 2.78 \text{ Hz}$
 $k = m(2\pi f)^2 = (207+160) \times 4\pi^2 \times (2.78)^2$
 $e = 160 \times (9.81) / k = 14.0 \text{ mm}$

7. [NYJC/Prelims 2014/P3/6]

- a. i. Force on structure by pivot, F



- ii. Loss in GPE = Gain in KE
 $mg(r - r \cos \theta) = 0.5mv^2$
 $9.81(0.400)(1 - \cos 5.00) = 0.5v^2$
 $v = 0.173 \text{ m s}^{-1}$

- iii. $F - W = m \frac{v^2}{r}$
 $F = 0.500 \left(\frac{0.1728^2}{0.400} + 9.81 \right)$
 $F = 4.94 \text{ N}$

By Newton's third law, the force exerted on the structure by the pivot is equal in magnitude and opposite in direction to the force exerted on the pivot by the structure. Hence force on pivot is 4.94 N.

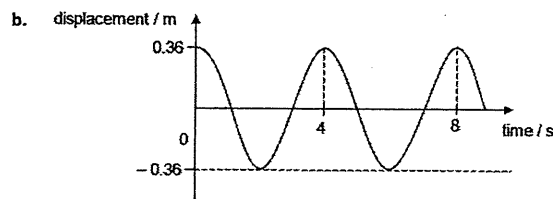
- iv. Not always vertical. This is because as the structure is oscillating (at any instant between Fig. 7.1 and Fig. 7.2), there is a centripetal acceleration toward O, which is diagonal. This suggests that the resultant force needs to have a horizontal component. As weight of the structure is always vertically downward, the only force that can provide the horizontal component can only be from the pivot.
- b. i. In phase means that the shadows of the peg and structure are always synchronised. The shadows always move to the same extreme location at the same time.
- ii. See definitions in lecture notes.
- iii. The shadow of the peg will oscillate faster than that of the structure, and the two shadows will not be in phase anymore. The phase difference will increase with time.
- iv. Since the peg is $1/3$ revolution per minute faster than the structure, this means that the peg will cover $1/3$ revolution more every one minute. To be in phase again, the peg needs to cover exactly one revolution more than the structure. Hence time required $= 1 \div \frac{1}{3} = 3 \text{ min} = 180 \text{ s}$
- v. Number of revolutions $= 3 \times 47 + 1 = 142$ revolutions

- vi. There is no loss in energy of the structure throughout the motion and hence the period/angular speed of the structure remains constant.
The turntable is able to maintain its angular speed/period due to the motor.

8. [PJC/Prelims 2014/P3/2]

- a. i. From the expression, the acceleration of the block is directly proportional to its displacement from its equilibrium position, and the acceleration is always directed towards that position as indicated by the negative sign. Thus the block is performing S.H.M. as any oscillatory motion that meets the above 2 conditions are oscillating with simple harmonic motion.

- ii. 1. From graph, $T = 4 \text{ s} \Rightarrow f = \frac{1}{T} = \frac{1}{4} = 0.25 \text{ Hz}$
2. Comparing $a = -\omega^2 x$ and $a = -\frac{28}{m}x$,
 $\omega^2 = \frac{28}{m} \Rightarrow 4\pi^2 f^2 = \frac{28}{m} \Rightarrow m = 11.3 \text{ kg}$
3. Max. k.e. $= \frac{1}{2} m \omega^2 x_0^2 = 1.8 \text{ J} \Rightarrow x_0 = 0.36 \text{ m}$



9. [VHS/Prelims 2014/P3/2]

- a. Resonance occurs when a system responds at maximum amplitude to an external driving force. This occurs when the driving frequency of external driving force equals to natural frequency of the system.
- b. i. $x = 0.18 \sin(3.5t)$
OR $x = 0.18 \cos(3.5t)$ where $\omega = (2\pi)(0.56) = 3.5$
- ii. $a = -\omega^2 x = -(3.5)^2 (0.080) = -0.98 \text{ m s}^{-2}$
- iii. $E_{k, \max} = \frac{1}{2} m v_{\max}^2 = \frac{1}{2} (1020) [(0.18)(3.5)]^2 = 202 \text{ J}$
- iv. Sine-squared graph, or cosine-squared graph with amplitude = 202 J and period = 1.8 s

10. [SAJC/Prelims 2014/P3/6(a),(b)]

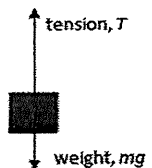
- a. i. 8.0 cm
- ii. $2\pi f = 220 \Rightarrow f = 35 \text{ Hz}$
- iii. line drawn mid-way between AB and CD (allow $\pm 2 \text{ mm}$)
- iv. $v = \omega x_0 = 220 \times 4.0 = 880 \text{ cm s}^{-1}$

- b. i. $d = -4.0 \cos(120^\circ) = 2.0 \text{ cm}$
 ii. $v = \sqrt{x_0^2 - x^2} = 220 \times \sqrt{4.0^2 - 2.0^2} = 760 \text{ cm s}^{-1}$
 iii. 1. line drawn 3/4 above AB
 2. arrow pointing upwards

11. [TJC/Prelims 2014/P3/2]

- a. See definitions in lecture notes.
 b. $v_{\max} = \omega x_0 = \frac{2\pi}{T} x_0 = \frac{2\pi}{0.80} \times 10.4 = 81.7 \text{ cm s}^{-1}$
 c. i. $mg - N = ma$
 $mg - 0 = ma$
 $a = g$
 $-\omega^2 x = g$
 $\left(\frac{2\pi}{0.80}\right)^2 x = 9.81$
 $x = -0.16 \text{ m}$
 From the graph, time = 0.52 s
 ii. At time = 0.2 s (at the amplitude point)

12. [TPJC/Prelims 2014/P3/2]

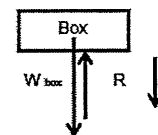
- a. When the object is moving towards the equilibrium position, its acceleration and velocity are in the same direction. When the object is moving away from the equilibrium position, its acceleration and velocity are in opposite direction.
 b. $T = kL$
 c. i. 
 ii. Newton's 1st Law: at equilibrium, $mg = kL$
 Newton's 2nd Law: $F_{\text{net}} = ma$
 $mg - k(x + L) = ma$
 $kL - k(x + L) = ma$
 $a = -(k/m)x$
 iii. The expression (k/m) is made up of constants, hence implies that the acceleration a of the mass is proportional to its displacement x .
 The negative sign in the expression $a = -(k/m)x$ also indicates that the acceleration is always in the opposite direction to the displacement.
 Both these satisfy the conditions for simple harmonic motion.

13. [VJC/Prelims 2014/P2/1]

- a. From 2nd graph, $v_0 = \omega x_0 = \frac{2\pi}{T} x_0$
 $T = \frac{2\pi x_0}{v_0} = \frac{2\pi \times 0.45}{1.2} = 2.4 \text{ s}$
 b. From 1st graph, $F_{\max} = m\omega^2 x_0$
 $m = \frac{F_{\max}}{\left(\frac{2\pi}{T}\right)^2 x_0} = \frac{0.63}{\left(\frac{2\pi}{2.36}\right)^2 \times 0.45} = 0.20 \text{ kg}$

14. [YJC/Prelims 2014/P3/3]

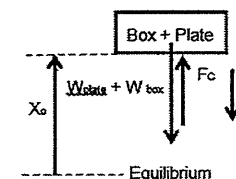
- a. $W_{\text{box}} + W_{\text{plate}} - F_c = 0$
 $1.0 \times 9.81 + 0.5 \times 9.81 - 150x = 0$
 $x = 0.0981 \text{ m}$
 b. $W_{\text{box}} - R = m_{\text{box}} a$
 $m_{\text{box}} g - 0 = m_{\text{box}} a$
 $a = g$



$$W_{\text{plate}} + W_{\text{box}} - F_c = (m_{\text{box}} + m_{\text{plate}}) a$$

$$0.5 \times 9.81 + 1.0 \times 9.81 - 150 \times (0.0981 - x_0) = (0.5 + 1.0) \times 9.81$$

$$x_0 = 0.0981 \text{ m}$$



- c. i. $KE_{\max} = \frac{1}{2} m x_0^2 \omega^2$
 $0.125 = \frac{1}{2} \times 1.0 \times 0.05^2 \times \left(\frac{2\pi}{T}\right)^2$
 $T = 0.628 \text{ s}$
 OR $T = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{1.0}{150}} = 0.628 \text{ s}$
 ii. Total energy = K.E. + G.P.E. = 0.045 + (1)(9.81)(0.01) = 0.143 J

